

Shape Up or Ship Out: Causal Effects of Performance Standards in Higher Education

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Abstract

Many higher education programs dismiss students who perform below some academic standard. Policymakers often justify these rules as a way to incentivize higher performance and redirect low-performing students toward more suitable career paths. Exploiting the national rollout of performance standards across Dutch bachelor programs, I provide the first comprehensive evidence on their effectiveness. Although the performance standards create strong incentives to increase effort and redirect many low-performing students toward alternative programs, they slightly reduce long-run degree attainment, do not shorten time in education, and have no measurable earnings effect. The results cast doubt on the effectiveness of performance-based dismissal policies.

Keywords: higher education, performance standards, dismissal policy

JEL Codes: I21, I23, I28

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1 Introduction

Only two in five students in OECD countries who begin a bachelor’s program complete it on time, while the rest either drop out or graduate with delay (OECD (2025)). This imposes substantial costs on both students and governments. For students, each additional year in education delays labor market entry and raises tuition expenses, while leaving without a degree can reduce lifetime earnings (Ost et al. (2018)). Governments bear much of the costs through subsidized tuition and lose potential tax revenue when students spend fewer years in the labor market. Across OECD countries, public and private spending on tertiary education averages about \$23,000 per student per year (OECD (2025)). Extended or incomplete studies thus create a substantial fiscal burden, even before accounting for lost output.

To limit these costs, many institutions worldwide use performance standards that dismiss students who perform below minimum academic requirements.¹ Policymakers view such performance standards as tools to raise graduation rates and reduce time in education through two main channels. First, they may increase academic achievement by attaching a penalty to low performance. Second, they may prevent late dropout or prolonged completion times by redirecting students unlikely to succeed toward alternative career paths at an early stage. Despite their widespread use, evidence on the effectiveness of performance standards is limited. Prior work has focused on students narrowly passing or failing performance standards, but shed little light on how students would perform in the absence of a performance standard. It therefore remains unclear whether the presence of performance standards improves performance, matching, and ultimately aggregate education or labor market outcomes.

This is the first study to provide a comprehensive evaluation of the effects of implementing performance standards. I exploit the staggered introduction of performance standards in 350 university bachelor programs in the Netherlands between 1995 and 2014. These performance

¹For instance, many higher education institutions worldwide dismiss students who score below certain GPA thresholds (e.g., US, Canada, Australia, New Zealand, Mexico, South Korea, Singapore, India), fail the same course multiple times (e.g., Germany, Switzerland, Denmark, Russia, and Austria), or fail a specific number of courses (e.g., the Netherlands, Indonesia, and South Africa).

standards require students to pass a specific number of courses by the end of their first year, and failure results in removal from the program. The requirements are strong: on average, one in five students is dismissed.² This means that the risk of dismissal extends beyond the lowest ability students. Combining a difference-in-differences design with administrative data covering over 700,000 students allows me to study the causal effect of these strict performance standards on both short- and long-term outcomes.

This approach identifies the average effect of introducing performance standards on all students in treated programs. Unlike approaches that focus on students at the margin of failing, the estimates capture the combined effects on all potentially affected students. This includes lower-ability students, who may exert more effort to remain enrolled or who switch career paths when dismissed, and higher-ability students, whose peer environment and effective class size change as weaker students exit. The results thus speak directly to policymakers seeking to assess the aggregate impact of performance standards.

I find that performance standards substantially alter student progression in the initial program. They increase dropout after the first year by 33 percent, corresponding to 7.6 percent of an average cohort dropping out who would otherwise have continued.³ The completion rate of all initially enrolled students decreases by 3.7 percentage points, which suggests that about half ($3.7/7.6$) of the additional dropouts would graduate in their initial program in the absence of the policy. The average enrollment duration of all initially enrolled students falls by 3 months, while the average completion time of graduates is unaffected. This indicates that the shorter enrollments are driven by earlier exit of dismissed students, rather than faster progress of non-dismissed students.

The performance standards do not improve long-run outcomes. Although their goal is to improve graduation, I find that instead performance standards slightly reduce degree

²This is estimated by the Dutch Ministry of Education through a survey (Ministerie van Onderwijs, Cultuur en Wetenschap (2010)). As explained in greater detail in section 3, I only observe dropout status, which can be voluntary or due to dismissal.

³The total share of students with insufficient credits is about 20 percent. This implies that over half of dismissed students ($12.4/20$) would also drop out in the absence of the standard.

attainment by 1.6 percentage points.⁴ Moreover, they do not reduce total time in higher education, another primary policy goal, nor do they affect employment or earnings up to twelve years after enrollment. The estimates are precise and rule out economically meaningful effects. I also find no evidence of positive effects in specific fields or cohorts, across different performance thresholds, in programs with many late dropouts, or among students with high ex-ante risk of failure.

A potential threat to the research design is that prospective students avoid programs with performance standards. I find no clear evidence of such deterrence. Enrollment numbers remain stable on average and across adoption cohorts or fields of study. The composition of new students by gender, socioeconomic background, and ability is also unaffected by the introduction of performance standards. These findings suggest that my results are unlikely driven by differential selection of incoming students.

Finally, I explore unintended negative consequences of performance standards. Beyond administrative burdens for program administrators, they may reduce student welfare by increasing pressure to perform or by removing students' potentially preferred option to continue in their initial program. To gauge this disutility, I presented 333 first-year students with hypothetical choice scenarios that trade off monetary gifts against immediate removal of the policy. More than 80 percent were willing to forgo some gifts, with 26 percent willing to forgo over €1,700.⁵ This suggests that first-year students experience disutility from performance standards. Using administrative data, I also test whether the policy increased antidepressant prescriptions, but find no evidence in support of this.

The results raise concerns about the effectiveness of similar performance-based dismissal policies in less forgiving contexts than the Netherlands. Dutch higher education is char-

⁴The Association of Dutch Universities states that the objectives of the performance standards are to 'reduce the high dropout rate in bachelor education, shorten the average study duration, and help students find a suitable program as quickly as possible' (Universiteiten van Nederland (2023)).

⁵For reference, the average monthly income of students is estimated at €943 (Groen and Houtsma (2021)) and the annual tuition fee is €2,530. Because the choices are hypothetical and unincentivized, the reported willingness to forgo gifts may overstate true willingness to pay, and should be interpreted as directional evidence of disutility rather than a precise welfare estimate.

acterized by relatively low tuition fees, minimal selective admission, and a high density of similarly ranked institutions. These conditions enable most dismissed students to re-enroll in alternative programs. In contrast, in countries with higher tuition, stricter admission, or fewer nearby options, dismissed students may be more likely to leave higher education entirely. Consistent with this, Ost et al. (2018) show that in the United States, students who narrowly fail a dismissal threshold often exit college and experience large earnings losses. In such contexts, performance gains among non-dismissed students would have to be substantial to offset the adverse consequences for those who are dismissed.

This paper adds to several strands of literature. Most closely related are studies on the effects of academic probation, which issues a formal warning to low-performing students and leads to dismissal if performance does not improve. Prior work finds that academic probation improves short-run academic performance for students near the probation threshold, but also increases dropout and induces shifts toward easier courses (Lindo et al. (2010), Fletcher and Tokmouline (2018), Casey et al. (2018), Wright (2020)).⁶ While these studies offer valuable insights, they leave two key questions for policymakers: (i) how would students perform in the absence of any performance standard, and (ii) how do performance standards affect the average rather than the marginal student.⁷ These questions matter because the presence of performance standards may alter effort and performance, and because effects on low- or high-performing students likely differ from those on students near the threshold. This paper addresses these gaps by testing, for the first time, whether the presence of performance standards improves average educational or labor market outcomes of all affected students.⁸

⁶Relatedly, Albert and Wozny (2024) study effects of more intensive academic probation which includes mandatory study time, and Canaan et al. (2023) and Ost et al. (2018) study effects of group coaching and academic dismissal for students already on probation.

⁷To my knowledge, only two papers have considered these aspects before, and they also focus on performance standards in the Netherlands (Arnold (2015), Sneyers and De Witte (2017)). Both consider only the share of students graduating within four years. Using more detailed administrative data, this paper examines whether total graduation rates, study duration, or labor market outcomes are affected.

⁸This distinction is similar to the literature on high-stakes exit exams, where some papers use regression discontinuity designs to study the effects of failing exit exams, and others use difference-in-differences designs to study the effects of the presence of (high-stakes) exit exams (e.g., Ou (2010), Clark and Martorell (2014), Caves and Balestra (2018), ter Meulen (2023), Fidjeland (2023)). A similar distinction arises in the literature on school accountability systems (Figlio and Loeb (2011)).

Second, the findings contribute to the literature on the optimal design of performance incentives in education more generally. Previous papers often consider relatively small-scale interventions, short-run outcomes, and (for ethical reasons) positive incentives (e.g., Angrist and Lavy (2009), Angrist et al. (2009), Leuven et al. (2010), Leuven et al. (2011), Rodríguez-Planas (2012), Bursztyn and Jensen (2015), Lavecchia et al. (2016), Papay et al. (2016), Clark et al. (2020), Hvidman and Sievertsen (2021)). However, results from small-scale interventions can differ greatly from scaled-up policy implementations (Al-Ubaydli et al. (2017)), negative incentives can have different effects than positive incentives due to asymmetric behavioral responses (Levitt et al. (2016)), and short- and long-run effects of interventions targeting human capital are often different (Almond et al. (2018), Aizer et al. (2024)). This paper contributes by providing unique evidence on the *long-run* effects of a *large-scale* implementation of a *negative* performance incentive.

Lastly, this paper relates to the literature on the efficacy of various selection mechanisms in higher education. A growing body of work studies selection *at the gate*: how and whether institutions should use (standardized) test scores, admission exams, interviews, CV screening, or affirmative action to efficiently allocate students to programs (Bettinger et al. (2013), Smith et al. (2015), Rothstein (2022), Bleemer (2022), Gandil and Leuven (2022), Chetty et al. (2026), Sacerdote et al. (Forthcoming)). A central concern is that pre-enrollment characteristics are noisy predictors of program-specific success, and that selection on these criteria may deter or disadvantage particular groups. This paper provides new insights on the efficacy of a complementary (or alternative) selection mechanism that operates *after the gate*: selecting students based on directly observed in-program performance. This provides a new perspective on how much programs learn about student potential from direct observation, and whether that information translates into accurate selection decisions.

This paper proceeds as follows. Section 2 provides institutional context. Section 3 presents the data. Section 4 discusses identification. Section 5 presents the main results. Section 6 explores effects on student wellbeing. Section 7 concludes.

2 Institutional context

Higher Education in the Netherlands. Higher education is offered by two types of institutions: research universities and universities of applied sciences. This paper focuses exclusively on bachelor programs at all 13 research universities. Bachelor programs offer specialized undergraduate education throughout the entire program, unlike, for example, in the United States, where students specialize later. All programs have a fixed curriculum in the first year, so strategic course taking in response to performance standards is not possible. Eligibility for these programs requires a high school diploma from the preparatory academic track or completion of the first-year curriculum at a university of applied sciences.⁹

Dutch university education possesses two distinctive features relevant to this paper. First, the majority of bachelor programs are required to admit all eligible applicants. Only a small number of programs are oversubscribed, in which case students are selected based on lotteries or additional information from CVs, interviews, and motivation letters. Hence, conditional on eligibility, admission is largely non-selective, and students who drop out are generally free to enroll in other programs. Second, universities in the Netherlands receive public funding, staff salaries are determined through collective agreements, and tuition fees for bachelor programs are standardized by law. As a result of the broad accessibility for students across programs, and the uniformity in terms of university funding and tuition fees, Dutch universities are considered to be qualitatively very similar.¹⁰

The bachelor programs use the European Credit Transfer System (ECTS).¹¹ The pro-

⁹For reference, in 2018, 19 percent of secondary school students in the Netherlands completed this track. Students from international backgrounds must present qualifications equivalent to the Dutch preparatory academic track to gain admission. Admission to some STEM programs further requires completion of designated STEM courses in secondary school.

¹⁰Avdeev et al. (2024) show that the variation in Dutch university rankings in the Times Higher Education ranking from 2023 is much lower than that of universities in the US, Sweden, Croatia, and Chile.

¹¹University programs have been split into bachelor and master programs since the Bologna reform in 2002. Before the reform, programs were undivided, and upon completion of a program, one was granted the equivalent of today's master's degree. The ECTS was also implemented as part of this reform. Before 2002, programs used a similar metric to assign credits to passed courses. While I also include undivided programs in my analysis, 85% of the programs introduced the performance standard after 2003 and therefore my results mostly apply to bachelor programs (rather than undivided programs).

grams consist of a combination of courses and a thesis, amounting to a total of 180 credits, which are awarded upon passing courses or the thesis. Students are expected to achieve 60 credits each year. However, in practice, students have the flexibility to adjust their course load or they may fail certain courses, causing delays in their study completion. Apart from the performance standards and personal costs and benefits, students faced few incentives to perform well or graduate on time. The main external incentive during the study period was that most students received a government grant for higher education that required graduation from any program within ten years.¹²

The Performance Standards. In the early 1990s, universities struggled with high dropout rates and long study durations, and this was particularly salient among students who performed poorly in their first year. This raised concerns among policymakers about the efficiency of university education. To address these issues, the government introduced a 1993 law requiring universities to advise all first-year students on whether to continue in their program. The government viewed such advice as a promising instrument to improve effort during the first year and redirect students unlikely to succeed toward more suitable fields of study. Although all programs were required to issue an advice, universities retained discretion over whether it was binding.¹³ Under a binding regime, known as the Binding Study Advice (BSA), students who receive a negative advice are dismissed from their program. Dismissed students remain eligible to enroll in other programs, either within the same institution or elsewhere. Credits for closely related courses completed in the initial program may be transferred to the new program, but recognition is discretionary and depends on the rules of the receiving program.¹⁴

¹²Students who failed to graduate within ten years had to repay the grant. Between 1993 and 1996, the grant was conditional on yearly performance.

¹³The estimates therefore capture the effects of having a binding rather than a non-binding study advice. The regulatory framework is specified in the Higher Education and Scientific Research Act (in Dutch: Wet op het Hoger Onderwijs en Wetenschappelijk Onderzoek, WHW) of 1993.

¹⁴Information on how often programs allow credit transfer is unavailable. Several universities explicitly prohibit the transfer of credits for first year courses. Because most dismissed students in Section 5 enroll in substantively different curricula, credit transfer are likely limited for this group.

When universities implement the BSA, they must adhere to specific requirements. First, they must establish clear criteria for issuing positive advice. This must include a minimum number of credits students must earn by the end of their first year. On average, this threshold requires students to pass 65% of their courses, though there is some variation among programs. Some programs may also have additional requirements, such as mandatory completion of specific courses or policies that allow students to offset failed courses with high grades in others. Second, the advice must be provided at the end of the first year, and programs are required to inform students in advance if they are at risk of receiving negative advice. Finally, universities must consider personal circumstances when making these decisions, and students have the right to contest negative advice.¹⁵

3 Data

Sample. I use administrative data from Statistics Netherlands covering the universe of students enrolled at Dutch universities. The sample includes all students who first enrolled in a bachelor program at a research university between 1991 and 2014.¹⁶ Each observation in my dataset represents a student and his/her initial enrollment. If a student later switches programs and re-enrolls as a first-year student elsewhere, they are not recorded as a new observation. This approach defines the initial enrollment as the start of the treatment period, treating any subsequent changes as outcomes. I exclude students who were above age 30 at the start of their studies (1%) and students in programs that (i) never implemented a performance standard or did not implement a performance standard by 2014 (2.9%), (ii) have a performance standard from the start (0.9%), or (iii) which ceased to exist before 2014

¹⁵In practice, about 4% of students appeal a negative binding study advice, and roughly 20% of them are successful (Ministerie van Onderwijs, Cultuur en Wetenschap (2010)). In addition, Vooren et al. (2024) show that some students continue their studies despite not meeting the minimum course requirements, suggesting that the rules are waived for some students on the basis of personal circumstances. Official figures for this group are, however, not available.

¹⁶Graduates from Universities of Applied Sciences often enroll in short pre-master's programs at research universities to qualify for a master's degree. However, the data do not distinguish between bachelor's and pre-master's enrollments. I therefore exclude all students who already hold a bachelor's degree from a University of Applied Sciences to ensure that the sample excludes pre-master students.

(1.5%). Following these selection criteria, my sample covers 707,932 students across 350 distinct bachelor programs. Whenever I refer to a program, I refer to a specific bachelor’s program at a specific institution. Below, I describe the construction of the key variables and present descriptive statistics of all students from the core sample.

Student characteristics. Table 1 shows that students are balanced by gender. Most enroll shortly after secondary school at about age 19. International students and those with prior enrollment at a University of Applied Sciences form 23 percent of the population. Using population and tax registers, I link students to their parents and measure the father’s income rank.¹⁷ Since not all students, especially international students, can be linked to their parents, this variable is available for 76 percent of students. Average paternal income is high compared to the national mean. For cohorts starting after 2006, I also observe students’ average secondary school grade in the final year. This measure is standardized across all other individuals from the same cohort and in the same level of secondary education.

Education outcomes. I first construct an indicator for first-year dropout, defined as not re-enrolling in the same program after the first year. This is relevant because performance standards determine whether students are allowed to continue after the first year. Table 1 shows that 26 percent of students drop out from the initial program after the first year, which can be voluntarily or through dismissal if the program has a performance standard.

The main long-run outcomes are students’ degree attainment and time spent in education. Table 1 shows that only 59 percent of students complete their initial degree. Since 26 percent drop out after the first year, this implies that 15 percent drop out after the second year or later. Because this is a sizable group, I also examine in section 4 whether performance standards affect the timing of dropout. Among all initially enrolled students, 78 percent eventually graduate from a research university, and this share rises to 86 percent when including graduation from a University of Applied Sciences.¹⁸ Students spend on average

¹⁷Income is first observed in 2001. For cohorts starting after 2001, income is measured in the year of enrollment; for earlier cohorts, income is measured in 2001. The rank is defined relative to all same-age individuals in the Netherlands with observed income.

¹⁸I observe enrollments and graduations up to 2022. This allows me to track education for at least nine

Table 1: Descriptive Statistics

	Cohorts	Mean	SD	N
<i>Characteristics</i>				
Male	All	0.5	0.5	707,932
Age	All	19.52	1.63	707,932
Foreign	All	0.12	0.32	707,932
Previously enrolled in University of Applied Sciences	All	0.11	0.31	707,932
Income rank father	All	0.75	0.25	534,358
Secondary school GPA	2007-2014	0.11	1	216,547
<i>Education outcomes</i>				
Dropout after the first year	All	0.26	0.44	707,932
Complete initial program	All	0.59	0.49	707,932
Obtain research university degree	All	0.78	0.42	707,932
Obtain higher education degree	All	0.86	0.34	707,932
Years enrolled in initial program	All	3.49	2.35	707,932
Years enrolled in higher education	All	6.24	2.75	707,932
<i>Labor market outcomes 12 years post enrollment</i>				
Missing earnings records	1991-2011	0.13	0.42	593,101
Positive earnings (conditional on non-missing)	1991-2011	0.96	0.2	514,595
Annual earnings (conditional on positive earnings)	1991-2011	55387	37009	492,734
Program size	All	93.7	120.63	350

Notes: This table presents descriptive statistics for the main variables in the analysis. ‘Cohorts’ refers to the years for which I observe the corresponding outcome. The remaining columns report the mean and standard deviation of each outcome and the number of observations underlying it.

3.5 years in their initial program and 6.2 years in higher education.¹⁹

Earnings. To accurately capture long-run earnings trajectories, I measure earnings 12 years after the initial enrollment. This corresponds to an average age of 31. Earnings encompasses earnings from employment and entrepreneurship, and is based on tax-return

years after entry for the last cohort (2013), with longer follow-up for earlier ones. Among students who entered in 2012, only 1.9% of students were still enrolled after nine years and did not obtain a degree yet.

¹⁹Although enrollment durations are reported in years for ease of interpretation, the underlying data are recorded monthly, which allows for fine granularity. The time spent in higher education is counted from the start of the initial enrollment until the end of the last enrollment, including breaks like gap-years. The results are the same when I define the enrollment duration as the sum of the yearly enrollments.

statements. I adjust earnings to 2015 prices. Because earnings are observed until 2024, this information is only available for cohorts that enrolled before 2012. I do not observe tax-returns for 13 percent of this sample, which is mostly due to migration of foreign students. Among the group for whom I do have tax returns, a small share (4%) have zero or negative earnings. In the analysis, I show that having missing records or having non-positive earnings is not related to being enrolled in a program with a performance standard, suggesting that selection into observed and positive earnings is unlikely an issue.

Program-level outcomes. In the analysis, I collapse the individual-level data to the program level. Unless stated otherwise, the outcomes Y_{pt} are averaged over all students who enrolled in year t in program p . The total number of year-by-program observations is 7,555, which is less than 350 (the number of programs) $\times 24$ (the number of years) $= 8,400$ because some programs were established after 1991.

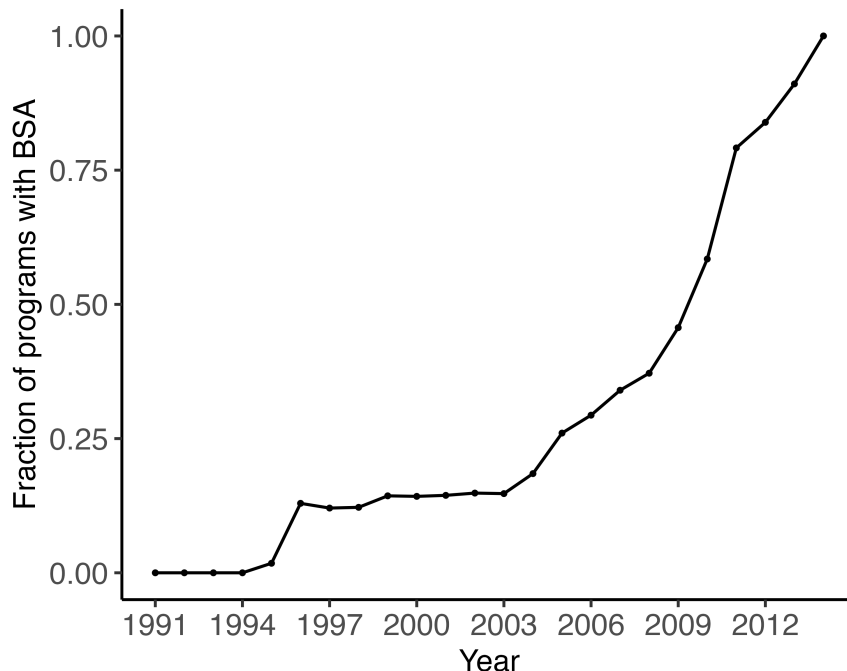
3.1 Implementation of the Performance Standard

I collected data on the introduction of the performance standards in programs from two sources. The Association of Dutch Universities provided me with records of programs between 1995 and 2012. For the years 2012 to 2014, I rely on annual reports from programs, faculties, or universities.

Figure 1 shows that the rate of adoption varied substantially across academic programs. Since 2003, additional programs have adopted a performance standard each year. By 2014, all programs in my sample have adopted the policy. Table A1 further documents the adoption at the institution level. Some universities introduced performance standards simultaneously across all programs, while others delegated the decision to faculties, creating within university variation. In total, there are 40 instances where a university adopted a performance standard for at least one of their programs.

The staggered adoption reflects divergent attitudes toward the policy over time and across institutions. The option to dismiss poor performing students introduced in 1993 sparked

Figure 1: Staggered rollout of the performance standard in bachelor programs



Note: This figure presents the share of programs with a BSA in each year between 1991 and 2014. The total number of programs in my sample is 350. The number of unique program in each year varies because each year some new programs are established.

significant debate among university and faculty boards, which had the authority to decide whether to adopt the BSA in their programs. Many academics and students strongly opposed dismissing students based solely on their first-year performance.²⁰ Critics also argued that the BSA would add administrative burdens, including an increase in appeals and procedural complexities. As a result, most universities were reluctant to directly implement the policy. At the same time, the Dutch government pressured universities to shorten the time students spend in programs and improve graduation rates, viewing the BSA as a promising strategy to achieve these national objectives. In 2006, the Minister of Education explicitly encouraged institutions to implement the BSA.²¹ This likely contributed to the faster and nearly universal adoption during and shortly after this period.

²⁰For example, the National Students' Union (LSVB) has consistently opposed the BSA since its inception, culminating in protests such as the 2009 occupation of the main building at the University of Groningen (Volkskrant, 2009)

²¹See 'Serious werk maken van bindend studieadvies', Volkskrant (2006).

Table A2 shows that the year of adoption is not related to program size, graduation rates, average enrollment duration, and field of study. This suggests that adoption timing is not strongly related to key program characteristics. Since programs that adopt performance standards at different times are similar along these dimensions, it is plausible that their outcome trends are also comparable. This assumption is central to the identification strategy, which I discuss in the next section.

4 Identification

I exploit the staggered rollout of performance standards to estimate their causal effect. This approach compares changes in student outcomes in programs that adopt a performance standard with changes in programs that adopt them later.

More specifically, I estimate event-study regressions of the form

$$Y_{pt} = \delta + \sum_{l=-5, l \neq -1}^2 \beta_l I[\tau_{pt} = l] + \alpha_p + \gamma_t + \epsilon_{pt}, \quad (1)$$

where Y_{pt} is the outcome corresponding to program p in year t . Unless otherwise stated, the outcomes are averages taken over all students who are initially enrolled in program p in year t . Moreover, τ_{pt} is the year relative to treatment, with $\tau_{pt} = 0$ in the first treatment period, and α_p and γ_t are program and year fixed effects. The regressions are weighted by the average program size in the pre-treatment periods. In this model, the effects of implementing performance standards l years ago are identified by β_l for $l \geq 0$. The estimates corresponding to $l < 0$ serve as placebo checks.

The key identifying assumption is that, in the absence of a performance standard, the outcomes of students in treated programs would evolve in parallel to those of students in untreated programs. There are multiple factors that support the plausibility of this assumption. First, Dutch bachelor programs are very homogenous. As explained in section 2, except for some oversubscribed programs, all bachelor programs have to admit all students.

Moreover, tuition fees are equal, universities are similarly funded, and geographical distances are relatively small. As a result of these features, the composition and quality of bachelor programs is very similar across universities. This makes it more likely that programs experience similar trends. Second, as performance standards are often uniformly implemented for entire faculties or universities, they are unlikely to be implemented in response to undesirable trends in specific programs. This is empirically supported by pre-trend estimates, which show that before implementation, treated and untreated programs evolve similarly.

Another key assumption is that programs' outcomes do not depend on the treatment of others. It is unlikely that students in programs with performance standards affect students in other programs directly because programs do not share courses in the first year. However, an indirect violation of this assumption arises when prospective students are deterred by the implementation of performance standards and substitute towards programs without performance standards. This results in a direct effect on the composition of students in treated programs, but it also affects the composition of untreated programs. I test for the importance of such selection effects in subsection 5.1, and find no clear evidence for compositional changes.

Methodological concerns. Recent papers have shown that standard two-way fixed effects regressions can yield misleading estimates, in particular when the fraction of treated units gets large over time (de Chaisemartin and D'Haultfoeuille (2020)). To overcome such issues, I use the Callaway and Sant'Anna (2021) doubly robust estimator. I also use their estimator of the aggregated Average Treatment effects on the Treated (ATT), which is a weighted average of the period-specific ATTs.

Other recently expressed concerns are related to model selection and inference. Raw data rarely exhibits parallel trends for treated and control units, and researchers commonly use different techniques, such as adjusting for covariates, to address this problem. However, conditioning the analysis on passing placebo checks induces pre-testing problems (Roth (2022)),

and, more generally, high degrees of freedom in specification choices can result in sizable replication problems (Menkveld et al. (2024)). To mitigate these concerns, I simply report estimates without conditioning on control variables. As a result, the ATT estimates are simple weighted averages of 2×2 difference-in-differences estimates between treated and not-yet-treated programs, with the weights equal to the programs' student shares.

Using this unconditional specification, I find at most mild deviations in pre-trends, and only for a limited number of variables. To check whether the estimates change when more carefully chosen control groups are used, I use Synthetic Difference in Differences (Arkhangelsky et al. (2021)). Unlike the baseline specification, which compares all treated programs to all not-yet-treated programs, this method matches treated programs to not-yet-treated programs with similar pre-treatment outcome trajectories. This approach automates the process of choosing appropriate control groups for each treated unit, all while retaining statistical guarantees. I find similar results using this approach.

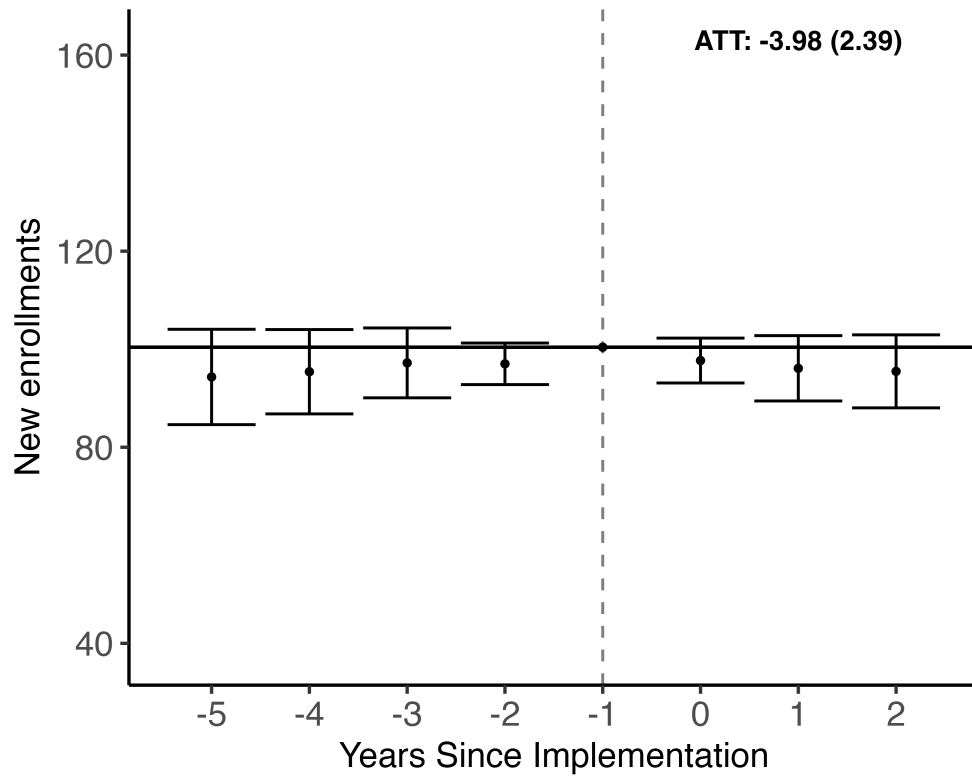
5 Results

5.1 Deterrence effects

In theory, imposing a performance standard is expected to discourage some students from enrolling for two reasons. First, as students cannot perfectly predict their performance in a new program, the performance standard introduces a risk of dismissal. This can deter students who are unsure about their ability to pass the threshold. Second, students who would otherwise exert too little effort to meet the standard are now confronted with the decision to raise their effort levels. For some of these students, increasing their efforts may not seem worthwhile, causing them to select another program.

The main outcome where a deterrence effect is expected to show up is the number of enrollments in the first year. However, Figure 2 shows that enrollment numbers decrease only marginally upon the introduction of performance standards. The ATT estimate itself

Figure 2: The effect of performance standards on new enrollments



Note: This figure presents event-study estimates of the effect of implementing performance standards on the number of first-year enrollments. The sample comprises all 7,555 yearly observations between 1991 and 2014 of the 350 programs. Unlike for the other specifications in this paper, the regression is not weighted by program size, as program size is the outcome of this regression. The ATT is based on the weighted average of the dynamic effects for the first three years after implementation. The graph is centered around the average program size in the last pre-treatment period. The error bars present 95% confidence intervals, adjusted for multiple hypothesis testing.

Table 2: The effect of performance standards on the composition of new students

	Income rank father	Male	Age	Previously enrolled at Uni- versity of Applied Science	Foreign	Secondary school GPA	Prob. of gradua- tion within four years
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ATT	0.001 (0.003)	-0.005 (0.005)	0.038 (0.029)	-0.001 (0.004)	-0.003 (0.006)	0.025 (0.018)	0.003 (0.004)
ATT ₋₁	-0.002 (0.003)	0.000 (0.005)	0.010 (0.018)	0.004 (0.004)	-0.005 (0.007)	-0.003 (0.016)	-0.002 (0.003)
ATT ₋₂	-0.001 (0.003)	-0.009 (0.005)	-0.008 (0.024)	0.002 (0.004)	-0.011 (0.007)	0.005 (0.021)	0.004 (0.004)
ATT ₋₃	-0.002 (0.003)	-0.009 (0.007)	0.017 (0.027)	0.007 (0.005)	-0.014 (0.009)		
ATT ₋₄	0.002 (0.003)	-0.007 (0.007)	-0.021 (0.031)	-0.002 (0.005)	-0.018 (0.010)		
Mean	0.754	0.484	19.431	0.104	0.133	0.103	0.513
SD	(0.041)	(0.233)	(0.491)	(0.086)	(0.145)	(0.398)	(0.089)
N	7,555	7,555	7,555	7,555	7,555	2,784	2,784

Notes: Each column presents estimates of the effect of implementing performance standards on a different outcome. Row 1 (ATT) presents the Average Treatment effect on the Treated, which is a weighted average of the dynamic effects for the first three periods after treatment. The next four rows present placebo estimates based on pre-treatment period differences. See Figure A4 for all event-study estimates. The last two rows report the mean and standard deviation of each outcome in the last pre-treatment period. Standard errors are in parentheses. (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$

is small (minus 0.033 standard deviations, p-value = 0.10), and the standard error rules out decreases in program size up to 0.074 standard deviations with 95 percent confidence. With synthetic difference-in-differences estimation, this reduces further to 0.056 standard deviations (Table A5). I find similar results when estimating results separately for large and small programs, when using the log of program size, or for particular fields or adoption cohorts (Figures A1, A2, and A3). Even in the earlier periods, when there were still many alternative programs without a performance standard, there is no clear drop in enrollments.

The minor reduction in program size may mask larger deterrence effects if performance standards deter some students while attracting others. In that case, the composition of

programs likely changes. To investigate this, I examine students' socioeconomic status, measured by their father's income, as well as their gender, age, previous enrollment in a University of Applied Sciences, nationality, and secondary school GPA. Since grades are only available since 2007, I provide pre-trend estimates for up to two years for this variable. Columns (1) to (6) of Table 2 show that these student characteristics do not meaningfully change upon the implementation of performance standards.

I next test whether treated programs attract students with higher predicted probabilities of on-time graduation based on their pre-enrollment characteristics. Specifically, I estimate each new student's likelihood of graduating within four years from their chosen program using the variables described above.²² This approach allows me to test whether students are selectively deterred from programs in which they are expected to perform worse. For instance, while the overall share of male students is unchanged, males may be disproportionately deterred from programs where they perform worse. Consistent with previous results, column (7) shows no evidence that implementing a performance standard increases the ex-ante quality of incoming students.

Finally, in Appendix C, I test for deterrence effects using an alternative identification strategy. I show visually and through regression results that whenever formerly popular programs among individuals who come from the same municipality implement a performance standard, then the number of prospective students from this municipality who enroll in programs with a performance standard increases proportionally. In other words, students from the same municipality consistently choose similar programs over time, and this is unaffected by the introduction of performance standards.

I conclude from this, and the evidence above, that any deterrence effect is, at most, small. This also implies that any effects on subsequent outcomes are unlikely explained by a differential selection of students once programs implement performance standards.

The absence of a stronger deterrence effect is somewhat surprising, given that about

²²Details of the prediction exercise are provided in Appendix B.

20 percent of students fail to meet the performance standards. Several explanations are possible. First, students may hold strong preferences for their chosen programs, leaving no preferred alternatives even when it includes the risk of dismissal. Second, students may be overconfident and therefore not perceive or strongly underestimate a risk of dismissal. Third, although programs were by law required to disclose the adoption of performance standards, some students may not have received or understood this information. Since the available data do not allow these mechanisms to be clearly distinguished—and because deterrence effects are not the main focus of this paper—I leave these questions for future work.

5.2 Effects in the initial program

First-year effects. By the end of the first year, students who did not pass the performance standard are not allowed to continue in their initial program.²³ Figure 3 shows that the introduction of performance standards greatly increases dropout rates after the first year. The figure is centered around the pre-treatment dropout rate of 23 percent and shows a direct and persistent increase by 7.6 percentage points (33 percent). This indicates that performance standards lead to the dismissal of a significant number of students who would otherwise have continued.²⁴

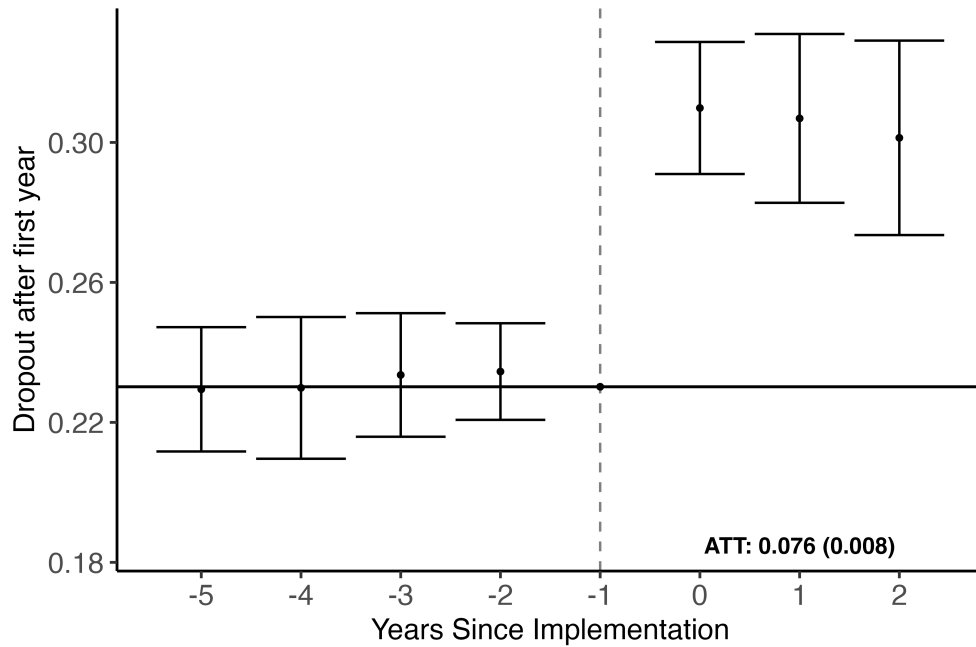
Table 3 shows what students do when they are dismissed. Columns 1 and 2 show that re-enrollment at another institution increases by 3.8 percentage points and dropout from higher education increases by 1.5 percentage points. This suggests that most dismissed students (5.3 of the 7.6 percentage point increase in dropout) make substantial changes in where or whether they study.

Dismissed students also make substantial changes in what they study. Enrollments to the same program at another institution increase by 1 percentage point (column 3), and

²³I do not observe the grades or the number of credits in the first year, preventing me from studying effects on short-term study performance.

²⁴Some dismissed students would also have dropped out in the absence of the performance standard. Given that the total share of dismissed students is estimated to be around 20 percent (Ministerie van Onderwijs, Cultuur en Wetenschap (2010)), this likely holds for over half of the dismissed students.

Figure 3: The effect of performance standards on dropout after the first year



Note: This figure presents event-study estimates of the effect of implementing performance standards on first-year dropout, which is an indicator for whether a student does not continue into the second year of the initially enrolled program. The sample comprises all 7,555 yearly observations between 1991 and 2014 of the 350 programs. The regression is weighted by the average program size in the pre-treatment periods. The ATT is based on the weighted average of the dynamic effects for the first three years after implementation. The graph is centered around the average dropout-rate in the last pre-treatment period. The error bars present 95% confidence intervals, adjusted for multiple hypothesis testing.

switching within the same field rises by 1.4 percentage points (column 4). This shows that among the dismissed students that do not leave higher education (6.1 p.p.), the majority make substantial changes in what they study: switching to a different field at a research university increases by 2.5 percentage points (column 5), and the remaining 1.2 percentage point enroll at a University of Applied Sciences (column 6).

Long-run effects. About half of the additional first-year dropouts would have dropped out later in the absence of the policy. Table 4 shows that performance standards reduce dropout after year two by 2.2 percentage points and dropout after year three or later by 1.7 percentage points. This suggests that performance standards result in the early dismissal of some students who would otherwise have dropped out later in the absence of the policy.

However, the increase in dropout after the first year is larger than the reduction in dropout after the second year and later. As a result, the overall completion rate in the initial program reduces by 3.7 percentage points (column 3). This reduction in the completion rate suggests that about half ($3.7/7.6$) of the additional policy-induced dropouts would have graduated in the absence of a performance standard. This estimate assumes that the graduation rate of non-dismissed students did not change. If performance standards raised their graduation rates—for example, by improving first-year performance—the implied share of policy-induced dropouts who would have graduated is even higher.

One of the primary benefits that programs mention for implementing performance standards is that they reduce the time that students spend in their program. Consistent with this, column 4 shows a reduction of 0.25 years (7.5 percent, or 0.39 SD) in the time students spend in their initial programs. This decline could arise from three channels: (i) dismissing students who would have dropped out later, (ii) dismissing students who would otherwise have stayed and graduated, or (iii) reducing the enrollment duration of students who always graduate. The last channel would reflect improved study performance among stayers.

However, I find no evidence for this last channel. Column 5 shows a precisely estimated

effect close to zero on average completion time. Because this outcome is conditional on graduation, and graduation itself is affected by performance standards, the estimate may reflect selection. Any bias is likely downward, however, because the students who do not graduate because of the performance standards are probably those with longer completion times in its absence. In other words, even though graduates in treated programs are likely positively selected *and* they faced strong incentives to perform during the first year, their average completion time does not reduce.

To summarize, performance standards substantially increase dropout after the first year. About half of these additional dropouts would have graduated without the policy. The average enrollment duration declines, reflecting earlier exits of students who would otherwise have remained enrolled, and in some cases graduated, rather than faster completion among stayers.

5.3 Long-term effects in higher education and the labor market

Higher education. Table 5 column 1 shows that implementing performance standards reduces graduation from any program at a research university by 1.6 percentage points. I find a slightly smaller estimate when I also include degrees from University of Applied Sciences in column 2. This reduction in graduation rates is lower than the reduction in the graduation rate from the initial program (3.7 percentage points), which implies that some of the students that dropped out from the initial program manage to graduate elsewhere. However, the goal of performance standards was to redirect students towards programs where they were *more* likely to graduate. The results suggest that performance standards do not achieve this goal.

Performance standards were also intended to reduce the time that students spend in higher education by redirecting students towards more suitable programs at an early stage. However, column 3 shows that the average time spent in higher education is unaffected. The estimate is precise and rules out enrollment duration reductions larger than 0.13 years

Table 3: The effect of performance standards on education choices after the first year

	Enroll at other institution	Leave higher education	Enroll in same program at research university	Enroll in same field but different program at research university	Enroll in other field at research university	Enroll in University of Applied Sciences
	(1)	(2)	(3)	(4)	(5)	(6)
ATT	0.038*** (0.005)	0.015* (0.006)	0.010*** (0.002)	0.014*** (0.003)	0.025*** (0.003)	0.012*** (0.003)
ATT ₋₁	0.009* (0.004)	0.002 (0.002)	0.001 (0.001)	-0.005 (0.003)	0.004 (0.003)	0.002 (0.003)
ATT ₋₂	0.006 (0.004)	-0.004 (0.002)	-0.001 (0.001)	-0.000 (0.003)	0.008* (0.004)	0.001 (0.003)
ATT ₋₃	0.001 (0.005)	0.001 (0.003)	0.001 (0.001)	-0.004 (0.004)	0.007 (0.004)	-0.005 (0.004)
ATT ₋₄	0.000 (0.004)	0.006 (0.004)	0.001 (0.001)	-0.007 (0.004)	0.002 (0.003)	-0.003 (0.004)
Mean	0.125	0.035	0.005	0.045	0.072	0.072
SD	(0.060)	(0.040)	(0.008)	(0.050)	(0.043)	(0.047)
N	7,555	7,555	7,555	7,555	7,555	7,555

Notes: Each column presents event-study estimates of the effect of implementing performance standards on a different outcome. Row 1 (ATT) presents the Average Treatment effect on the Treated, which is a weighted average of the dynamic effects for the first three periods after treatment. The next four rows present placebo estimates based on pre-treatment period differences. See Figure A5 for all event-study estimates. The next two rows report the mean and standard deviation of each outcome in the last pre-treatment period. Standard errors are in parentheses. (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$

Table 4: The effect of performance standards on outcomes in the initial program

	Dropout after year two	Dropout after year three or later	Complete initial program	Years enrolled in initial program	Completion time (conditional on graduation)
	(1)	(2)	(3)	(4)	(5)
ATT	-0.022*** (0.003)	-0.017*** (0.003)	-0.037*** (0.007)	-0.247*** (0.037)	-0.052 (0.031)
ATT ₋₁	0.002 (0.003)	0.005 (0.003)	-0.011* (0.005)	-0.025 (0.023)	0.001 (0.026)
ATT ₋₂	0.002 (0.004)	0.005 (0.004)	-0.011 (0.007)	-0.032 (0.031)	-0.025 (0.032)
ATT ₋₃	0.007 (0.004)	0.016** (0.006)	-0.022** (0.008)	0.025 (0.037)	0.036 (0.035)
ATT ₋₄	0.003 (0.004)	0.010* (0.005)	-0.013 (0.009)	0.027 (0.043)	0.019 (0.043)
Mean	0.072	0.061	0.637	3.293	4.261
SD	(0.040)	(0.047)	(0.124)	(0.628)	(0.890)
N	7,555	7,555	7,555	7,555	7,555

Notes: Each column presents event-study estimates of the effect of implementing performance standards on a different outcome. Row 1 (ATT) presents the Average Treatment effect on the Treated, which is a weighted average of the dynamic effects for the first three periods after treatment. The next four rows present placebo estimates based on pre-treatment period differences. See Figure A6 for all event-study estimates. The last two rows report the mean and standard deviation of each outcome in the last pre-treatment period. Standard errors are in parentheses. (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$)

(2 percent) with 95 percent confidence. This shows that although the average enrollment duration in the initial program is reduced, dismissed students often re-enroll elsewhere and spend as much time in higher education as they would have done otherwise. I also find no effects on the 90th percentile of the enrollment duration distribution (column 4), suggesting no visible impact even among the slowest students.

Labor market outcomes. I next study effects on post-study earnings. This dimension is relevant because the higher education outcomes above may not fully reflect students' human capital accumulation. For example, even when the effects on higher education attainment and time in higher education are small, students' labor market outcomes may have improved if they shifted towards more suitable careers or if their performance improved in a way that is not reflected in graduation rates and enrollment durations.

Since earnings twelve years after enrollment are observed only for earlier cohorts, I restrict the sample to students enrolling before 2012 (Table 1). I first verify that performance standards do not affect selection into observed earnings in Table A3.²⁵ The estimates are insignificant, indicating that selective observation of earnings is unlikely to confound the estimates.

Column 5 of Table 5 reports a precisely estimated null effect on average earnings, ruling out increases larger than 1.6 percent with 95 percent confidence. Table A3 shows no effects at the 10th or 90th percentiles, indicating that performance standards also do not affect within-program earnings inequality. Moreover, Table A4 documents no earnings effects in any year from enrollment through year twelve, suggesting that performance standards also do not alter earnings growth trajectories.

The estimates establish that performance standards do not improve average earnings across all potentially affected students. This includes dismissed students who fail to graduate because of the policy, dismissed students who graduate elsewhere, low-performing students

²⁵Table A3 also reports two supplementary checks. First, the effect on dropout rates is comparable in magnitude to the full sample, suggesting that those cohorts are similarly affected. Second, there is no effect on migration, ruling out differential attrition through geographic mobility.

Table 5: The effect of performance standards on long-term outcomes

	Obtain research university degree	Obtain higher education degree	Years enrolled in higher education	Years enrolled in higher education (90th quantile)	Annual earnings 12 years post enrollment (€)
	(1)	(2)	(3)	(4)	(5)
ATT	-0.016** (0.006)	-0.013* (0.006)	-0.061 (0.037)	-0.010 (0.061)	348 (357)
ATT ₋₁	-0.008* (0.004)	-0.007 (0.004)	0.010 (0.029)	0.055 (0.065)	-313 (306)
ATT ₋₂	0.003 (0.005)	0.004 (0.004)	0.004 (0.032)	0.019 (0.066)	481 (406)
ATT ₋₃	-0.004 (0.007)	-0.004 (0.006)	0.003 (0.039)	0.037 (0.072)	578 (522)
ATT ₋₄	-0.002 (0.006)	-0.004 (0.006)	0.006 (0.045)	0.115 (0.082)	127 (530)
Mean	0.804	0.885	6.239	8.980	63,565
SD	(0.081)	(0.063)	(0.800)	(1.078)	(14,436)
N	7,555	7,555	7,555	7,555	6,531

Notes: Each column presents event-study estimates of the effect of implementing performance standards on a different outcome. With the exception of column 4 and 5, the program-level outcomes are averaged over all students who initially enrolled in the respective program. The outcome in column 4 is the 90th quantile of the enrollment duration distribution, which is also taken over all initially enrolled students. Earnings in column 5 is taken over all initially enrolled students with observed and positive earnings. Row 1 (ATT) presents the Average Treatment effect on the Treated, which is a weighted average of the dynamic effects for the first three periods after treatment. The next four rows present placebo estimates based on pre-treatment period differences. See Figure A7 for all event-study estimates. The last two rows report the mean and standard deviation of each outcome in the last pre-treatment period. Standard errors are in parentheses. (***) $p < 0.001$, ** $p < 0.01$, * $p < 0.05$)

who remain in the program because they increased effort, and higher ability students exposed to changes in peer composition and reductions in program size. That this average effect is near zero despite a modest reduction in graduation rates does not imply that degrees carry no returns. The confidence interval on the earnings estimate does not rule out meaningful negative effects on the 1.6 percent of students who fail to graduate because of the policy. Moreover, opposing effects across the other affected groups may cancel in the aggregate, potentially masking negative effects on this specific subgroup.

Additional results. Table A5 shows that the main results are similar when using a Synthetic Difference-in-Differences approach, which is consistent with only minor deviations in pre-trends even without matching. Table A6 shows that the results also do not change when excluding international students, whose visa depends on continued enrollment and for whom dismissal is therefore more consequential.

Overall, the results suggest that performance standards do not meaningfully improve students' average education or labor market outcomes. The next section examines whether these average effects mask heterogeneous effects.

5.4 Heterogeneity

To limit the number of hypothesis tests and keep the results concise, I focus on two outcomes: (i) the dropout rate after the first year, to explore which groups are most impacted by performance standards in the short run, and (ii) the average enrollment duration per graduate. The latter is defined as the average number of years that all initially enrolled students are enrolled in higher education divided by their average graduation rate in higher education. The advantage of this index is that it summarizes long-run effects on two key outcomes into one measure, and it can be interpreted as a crude cost benefit ratio. If performance standards reduce enrollment durations (costs) or increases graduation rates (benefits), then this ratio should decrease. Figure 4 reports estimates for these outcomes for several subgroups.

Heterogeneity by student type. I start by exploring heterogeneity by gender because males perform worse in higher education and drop out more often. A plausible explanation is that males are less patient, leading them to underinvest in their studies (Castillo et al. (2011)). By imposing an incentive to perform early in the program, performance standards could therefore be more beneficial for men. Consistent with lower performance among men, Figure 4 shows that men are more likely to be dismissed. However, in the long run, there is no improvement in the average enrollment duration per graduate for men or women.

Second, I test whether performance standards differentially affect high-risk students, where high-risk students are defined as the 25 percent students in each program with the lowest predicted on-time graduation probability based on their pre-enrollment characteristics (as in Section 5.1). As expected, these students are more likely to be dismissed. However, also for this group, I find no long-run benefits.

Third, I test whether performance standards differentially affect low-income students, where low-income students are defined as the 30 percent of students in the core sample with the lowest paternal income rank. I find similar effects on both outcomes for low- and high-income students.

Heterogeneity by program type. I first consider programs with a high share of late dropouts, defined as programs with an above-median share of students dropping out after the second year or later in the pre-treatment periods. Since performance standards were intended to redirect unfit students into more suitable programs at an early stage, these programs offer the greatest scope for improvement. However, Figure 4 shows that while dismissals are indeed somewhat more frequent in these programs, there are no effects on long-run outcomes for this group neither. For programs with relatively few late dropouts, performance standards even slightly increase the average enrollment duration per graduate.

Second, I compare programs by threshold level. Below-median programs require first-year

students to obtain 30–39 credits (out of 60) to pass, while above-median programs require 40–48 credits.²⁶ Dropout effects are similar across both groups, and in the long-run the enrollment duration per graduate even increases in high-threshold programs.²⁷

Third, I consider heterogeneity by field of study. Although short-run effects vary across fields, the long-run impacts are also consistently small.

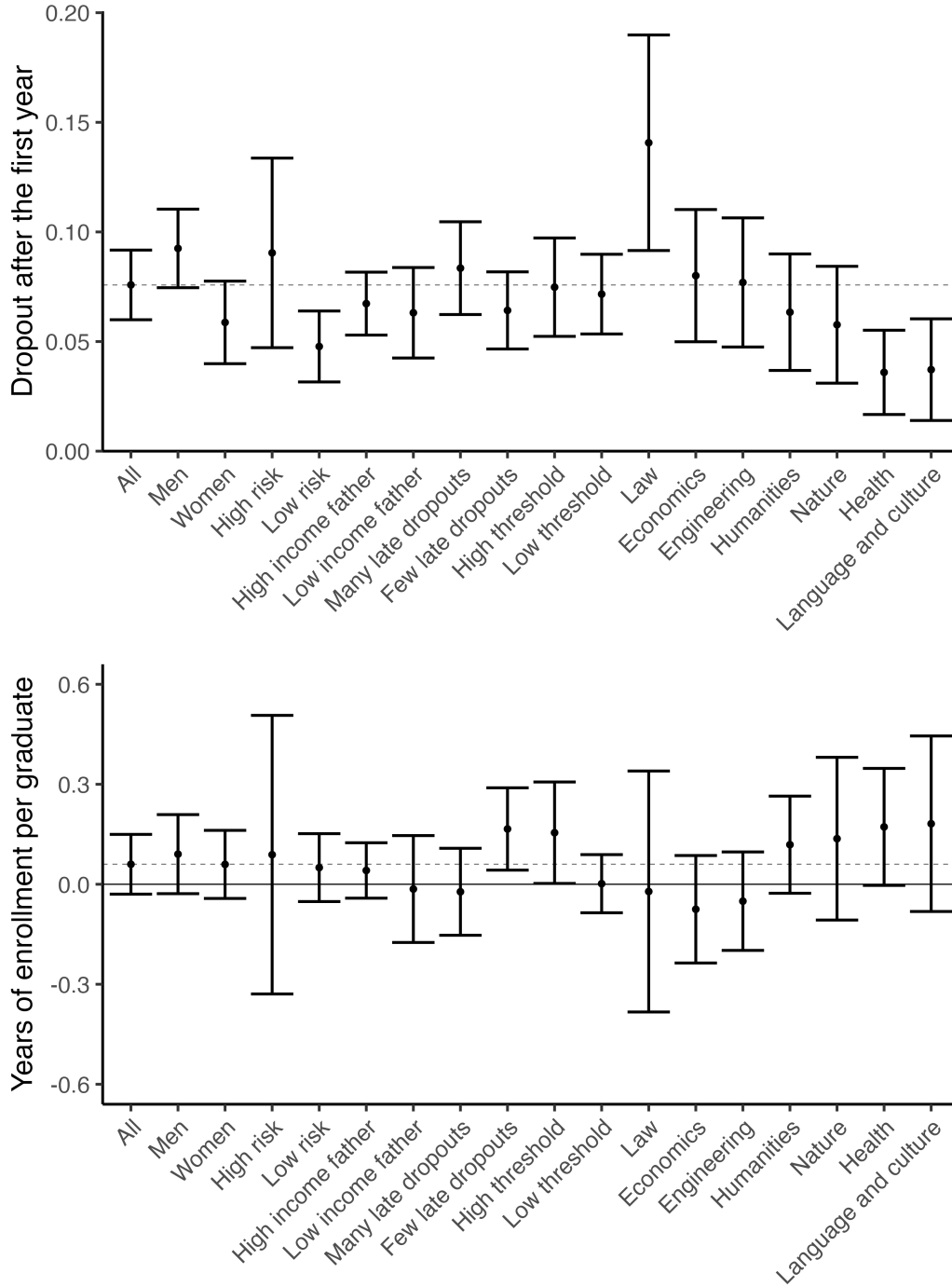
Heterogeneity over time. Figure A8 shows effects across adoption cohorts. I find that the dismissal rate is somewhat higher among the early adopters, and this also results in longer-run impacts. However, these effects correspond only to a small number of programs (15 percent), are mostly negative, and are quite imprecisely estimated. For the larger share of programs that implemented performance standards after 2003, the average enrollment duration per graduate is not affected.

To conclude, while the short-run effects vary somewhat between students, programs, or cohorts, the long-run effects are consistently small. Performance standards do not measurably improve the average enrollment duration per graduate for males or females, low- or high-risk students, low- or high-income students, programs with a high or low share of late dropouts, programs with high or low thresholds, programs in specific fields, and — with the exception of a small share of early adopters — the effects are consistently small across implementation cohorts.

²⁶These are the thresholds at implementation. Some programs later changed their thresholds.

²⁷Programs where students tend to obtain fewer credits in the first year, such as STEM programs, tend to set lower thresholds, which may explain why dropout effects are comparable in low-threshold programs.

Figure 4: Heterogeneous effects of performance standards



Note: This figure presents estimates of the effect of implementing performance standards on first-year dropout and the average enrollment duration in higher education per graduate for several subgroups. The estimates by student type are obtained by splitting the individual data by the respective type, collapse the data to the program level for each type separately, and apply difference-in-differences to each dataset separately. The estimates by program type are obtained as follows: (i) select all treated and untreated observations for programs from a specific type and select all untreated observations from all other programs, (ii) apply difference-in-differences to this subsample to obtain the type-specific ATT, and (iii) repeat this process for all other types. The dashed line is centered around the ATT for the full sample. The error bars present 95% confidence intervals.

6 The non-monetary costs of performance standards

6.1 Measuring (dis)utility with survey evidence

Performance-based dismissal policies restrict the failing students' choices by taking away the option to continue in their program. In the absence of spillovers or behavioral biases, reducing choice sets is unlikely to increase welfare. The intervention is not free of monetary costs either, because universities face administrative costs and appeal cases.

These costs can be justified when performance standards reduce costs imposed on others. For example, the goal was to reduce the time students spend in education, thereby reducing education subsidies and forgone earnings. However, the results indicate that performance standards do not improve students' time spent in education or labor market outcomes.

Another justification could be an increase in positive externalities (Chetty (2015)). For example, adolescents are known to disproportionately discount the future benefits of education (Bleemer and Zafar (2018)), inducing them to invest suboptimally in their studies. If that is the case for first-year bachelor students too, then the performance standard can be a welcome commitment device that gives students a more immediate incentive to increase their effort level (Clark et al. (2020)).

To test whether the performance standard is perceived as a distortionary intervention or a welcome commitment device, I surveyed 333 first-year students in two bachelor programs from a large university in the Netherlands. The survey is conducted in the first semester, when the students finished their first exams but have not received grades yet. They are asked five versions of the same hypothetical choice question where they trade off monetary gifts and the immediate removal of the performance standard.²⁸ I use a staircase method that enables me to narrow down the interval around the amount where each student is exactly indifferent between the gift and the removal of the performance standard.²⁹ Importantly, for

²⁸The exact phrasing of the question and other details of the survey are discussed in Appendix D.

²⁹Similar staircase methods are commonly used to calculate individuals' certainty equivalents or individual time discounting rates (Falk et al. (2018)).

students with a sufficiently low willingness to forgo gifts, I also present the option to *pay* a small amount of money to keep the performance standard in place. This option corresponds to students who appreciate the presence of a performance standard.

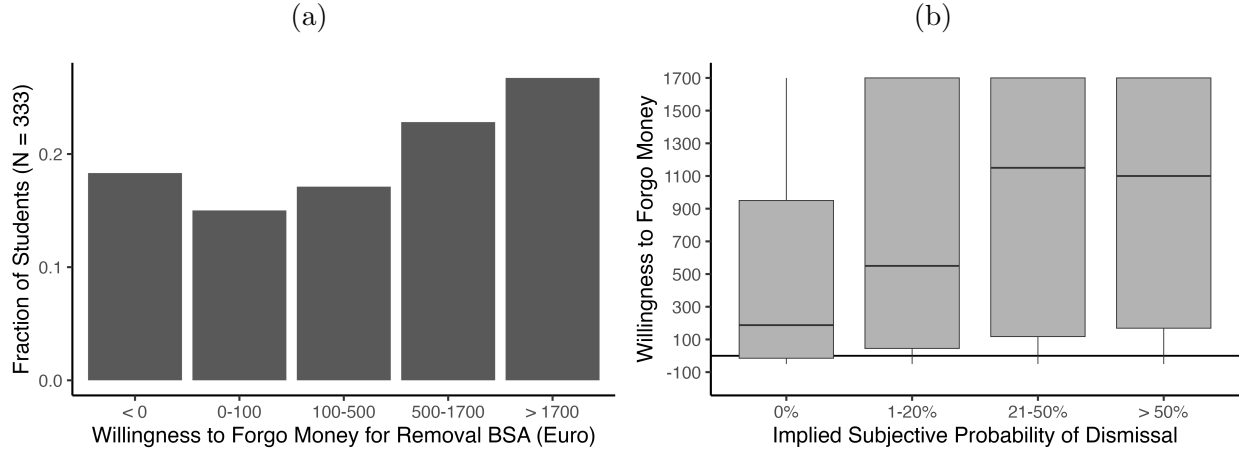
An advantage of the hypothetical choice approach is that it enables flexible estimation of the full distribution of preferences (Wiswall and Zafar (2018)). This allows me to detect both individuals who appreciate and who dislike the performance standard. Even if administrative data included measures of student utility or satisfaction—which it does not—it would remain difficult to capture such preference heterogeneity. A downside is that willingness to forgo money is hypothetical and elicited before students fully experience the long-run consequences of the policy. Students who dislike the performance standard in the first year may revise their views if they later believe it improved their performance or led to a better matched program. The results should therefore be interpreted as directional evidence of (dis)utility rather than precise estimates of welfare costs.

Figure 5(a) shows the distribution of students’ willingness to forgo money for the immediate removal of the performance standard. A minority (18 percent) would pay to retain the standard, suggesting that for some it serves as a commitment device. Yet a much larger share indicates a willingness to forgo money to remove it. For example, 24 percent would give up between 500 and 1700 euros, and 26 percent indicate the maximum of 1700 euros. This far exceeds the average monthly disposable income of students, estimated at 943 euros (Groen and Houtsma (2021)).

While the willingness to forgo gifts increases with the subjective risk of dismissal, even students who perceive little risk are often willing to forgo gifts. To measure this risk, I elicit students’ expectations about the number of course credits they will have by the end of the year and use these to construct the implied probability of dismissal.³⁰ Figure 5(b) shows boxplots of the willingness to forgo gifts against these subjective probabilities. The median

³⁰Specifically, instead of asking about their subjective likelihood of dismissal directly, I follow Manski (2004) and ask students to allocate 100 points over each possible number of credits they can obtain. I then define a student’s subjective probability of being dismissed by the implied probability that a student has fewer credits than necessary to pass the performance standard.

Figure 5: Willingness to forgo gifts for removal of the performance standard



Note: Figure (a) presents a bar chart of students' willingness to forgo gifts in return for the immediate removal of the performance standard. The willingness to forgo money is based on survey responses to five hypothetical choice questions, discussed in further detail in Appendix D. Figure (b) shows boxplots of the willingness to forgo money for four groups of students, sorted based on their implied subjective probability of dismissal. The boxes mark the interquartile range, the line inside marks the median, and whiskers extend to 1.5 times the interquartile range. The share of students in the bins is 37% (bin 0), 24% (bin 1-20), 19% (bin 21-50), and 20% (bin > 50).

willingness to forgo gifts increases steeply with dismissal risk, suggesting that students most likely to fail the standard experience the strongest disutility. Yet the median willingness remains positive in all four probability bins, including among students who are certain they will have sufficiently many credits. This indicates that the disutility from performance standards extends beyond those at risk of dismissal.³¹

6.2 Mental health

A specific way through which performance standards can reduce students' utility is if it unintentionally impacts their mental well-being. Student mental health in the Netherlands has shown a declining trend, and the binding study advice is repeatedly mentioned as a potential contributor (Dopmeijer et al. (2022)). Despite anecdotal evidence linking mental health concerns to performance standards, there exists no causal evidence on its effect.

To shed light on this, I examine the effect of performance standards on antidepressant

³¹A plausible explanation is that these students expect to meet the credit requirement if they apply sufficient effort, but would prefer to exert less effort than the performance standard demands.

prescriptions in two windows: (i) within two years of enrollment, capturing psychological pressure and short-run effects of dismissal, and (ii) five to eight years after enrollment, capturing potential career dissatisfaction among dismissed students. Prescription data are available from 2007, so the first outcome is restricted to students enrolling from 2007 onward, and the second to those enrolling from 2002.

I find no evidence that performance standards raise antidepressant use. Table A7 reports small, insignificant estimates, ruling out increases larger than 0.5 percentage points (18 percent) within two years and 0.7 percentage points (11 percent) five to eight years after enrollment with 95 percent confidence.

While these findings are a step towards a better understanding of mental health effects, they do not rule out negative effects on less extreme mental health outcomes. Performance standards may affect students' mental health in other ways that do not directly manifest in antidepressant usage.

7 Conclusion and discussion

This paper evaluates the introduction of strict performance standards in Dutch bachelor programs between 1995 and 2014. The performance standards were intended to improve student outcomes by incentivizing effort in the first year and redirecting students unlikely to succeed toward more suitable career paths. My results suggest that these objectives are largely not achieved.

In the short-run, performance standards result in the dismissal of many students from their preferred programs. In the long run, they slightly reduce degree attainment without shortening time spent in higher education, and they do not improve labor market outcomes. Survey evidence further indicates that performance standards cause disutility.

Taken together, the findings cast doubt on the effectiveness of performance-induced dismissal as a tool to improve student outcomes. The Dutch higher education system offers

relatively generous outside options for dismissed students, who can re-enroll in many alternative programs at low costs. Even in this environment, the policy fails to generate education or labor market benefits. This raises concerns about similar policies in less forgiving contexts. Where tuition is higher, admission constraints are tighter, or nearby alternatives are scarce, dismissal may push students out of higher education entirely, with potentially sizable earnings losses as a result (Ost et al. (2018)). In such settings, any gains among continuing students would need to be large to compensate for losses among those who are dismissed.

Lastly, the estimates capture average effects across all initially enrolled students and focus on long-run outcomes such as degree attainment, enrollment duration, and earnings. This approach provides important insights for policymakers on whether performance standards ultimately improve aggregate educational and labor market outcomes. However, it does not allow for a detailed examination of the mechanisms underlying the absence of long-run benefits. Because grades and credits are not observed, I cannot assess whether initial performance gains faded over time or distinguish effects between continuing and dismissed students. Some students who met the thresholds may have benefited in the long run, but such gains could be offset by losses among dismissed students. These mechanisms remain an important direction for future research.

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Appendix A: Supplementary Tables and Figures

Table A1: The adoption of performance standards by university

University	1995	1996	1999	2001	2002	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
1											3		1	13	2	
2			1				14									
3	5	33														
4												43				
5					4	2		1					1	1	25	12
6											13	1				
7											11					
8						2	3	11	17	8	1					
9							2			3		1	34	1		
1						8	6									
11			4	3		1	1				1	1				
12													37			
13																19

Notes: This table reports for each of the thirteen universities how many programs adopted a performance standard in each year.

Table A2: The relationship between program characteristics on and adoption timing of performance standards

	Year of adoption performance standard
log(program size)	-0.254 (0.325)
Graduation rate	1.735 (3.327)
Average enrollment duration	-0.976 (0.695)
Field = agriculture	6.355* (3.220)
Field = Life sciences	0.975 (3.121)
Field = Engineering	2.859 (3.196)
Field = Health	1.408 (3.148)
Field = Economics	-2.233 (3.172)
Field = Law	-2.437 (3.269)
Field = Humanities	0.185 (3.045)
Field = Language and culture	-0.765 (3.116)
N	350

Notes: This table presents results from a regression of the year of adoption of a performance standard on the log of the average program size, the graduation rate and average enrollment duration, and indicators for the field of study. These variables are averaged over all enrolled students in each program from the core sample. The left-out indicator for field of study corresponds to Education Sciences. (***) $p < 0.001$, ** $p < 0.01$, * $p < 0.05$)

Table A3: Supplementary results with the earnings-only sample (1991-2011)

	Dropout after first year	Missing earnings	Positive earnings	Out- migration	Earnings (10th quantile)	Earnings (90th quantile)
	(1)	(2)	(3)	(4)	(5)	(6)
ATT	0.089*** (0.008)	0.015 (0.008)	0.001 (0.002)	-0.000 (0.005)	66 (588)	693 (576)
ATT ₋₁	0.006 (0.006)	0.003 (0.005)	0.002 (0.003)	0.001 (0.004)	-1002 (694)	-464 (648)
ATT ₋₂	0.002 (0.008)	-0.011* (0.006)	0.003 (0.003)	-0.006 (0.005)	-44 (769)	289 (736)
ATT ₋₃	-0.002 (0.008)	-0.013 (0.010)	0.001 (0.002)	-0.009 (0.008)	739 (731)	-87 (861)
ATT ₋₄	-0.002 (0.008)	-0.007 (0.012)	0.001 (0.003)	-0.005 (0.009)	223 (708)	-284 (828)
Mean	0.234	0.153	0.958	0.127	30,232	93,134
SD	(0.090)	(0.155)	(0.028)	(0.110)	(12,330)	(20,119)
N	6,531	6,531	6,531	6,531	6,531	6,531

Notes: Each column presents event-study estimates of the effect of implementing performance standards on a different outcome. The sample corresponds to all program-year observations from 1991 to 2011. The outcomes in columns 1, 2, and 4 are averaged over all initially enrolled students. The outcome in column 3 is averaged over all individuals with non-missing earnings. The outcomes in columns 5 and 6 measure the 10th and 90th quantiles of the earnings distribution, which are also taken over all initially enrolled students. Row 1 (ATT) presents the Average Treatment effect on the Treated, which is a weighted average of the ATTs for three post-treatment periods. The next four rows present placebo estimates based on pre-treatment period differences. The last two rows report the mean and standard deviation of each outcome in the last pre-treatment period. Standard errors are in parentheses. (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$

Table A4: Effects on average earnings from 0 to 11 years since first enrollment

Average earnings												
ATT	-364	-165	-62	159	173	312	172	116	107	72	99	99
	(415)	(459)	(251)	(264)	(251)	(196)	(200)	(126)	(83)	(75)	(54)	(55)
Mean	57,169	51,891	44,252	35,787	26,044	16,465	9,984	6,595	4,937	3,923	3,090	2,620
SD	12,318	11,869	9,485	7,891	6,349	5,403	3,712	2,134	1,442	1,088	761	630
N	6,858	7,205	7,555	7,555	7,555	7,555	6,228	5,942	5,660	5,373	5,080	4,785
Share with missing or non-positive earnings												
ATT	0.00	0.01	0.01	0.01	0.00	0.01	-0.01	0.00	-0.01	-0.01	-0.01	0.00
	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Mean	0.18	0.18	0.18	0.18	0.20	0.22	0.23	0.24	0.26	0.27	0.29	0.27
SD	0.15	0.15	0.15	0.14	0.14	0.16	0.15	0.15	0.15	0.14	0.14	0.14
N	6,858	7,205	7,555	7,555	7,555	7,555	6,228	5,942	5,660	5,373	5,080	4,785
Years since first enrollment	11	10	9	8	7	6	5	4	3	2	1	0
Cohorts	1991-2012	1991-2013	All	All	All	All	1996-2014	1997-2014	1998-2014	1999-2014	2000-2014	2001-2014

Notes: This table presents estimates of the effect of implementing performance standards on average annual earnings (upper panel) and on the share of students with missing or non-positive earnings records (lower panel), measured at each year from first enrollment through eleven years after. The outcomes are averaged over all initially enrolled students; average earnings are conditional on having observed and positive earnings. The set of cohorts varies across columns because earnings are observed only between 2001 and 2024. Row 1 (ATT) reports the Average Treatment effect on the Treated, computed as a weighted average of the period-specific ATTs for the first three post-treatment periods. The bottom rows report the mean and standard deviation of each outcome in the last pre-treatment period. Standard errors are in parentheses. (***) $p < 0.001$, ** $p < 0.01$, * $p < 0.05$)

Table A5: Main results, using Synthetic difference-in-difference

	New enrollments	Dropout after first year	Complete initial program	Years enrolled in initial program
ATT	-3.390 (1.753)	0.070*** (0.011)	-0.026*** (0.006)	-0.237*** (0.046)
N	7,555	7,555	7,555	7,555
	Obtain research university degree	Obtain higher education degree	Years enrolled in higher education	Annual earnings 12 years post enrollment
ATT	-0.016** (0.005)	-0.013* (0.006)	-0.065 (0.043)	300 (784)
N	7,555	7,555	7,555	6,531

Notes: This table presents synthetic difference-in-difference estimates of the effect of performance standards on various outcomes. The estimates are obtained by applying synthetic difference-in-differences to each adoption cohort separately, and then aggregating their results (Arkhangelsky et al. (2021)). All results are based on the same samples as in the main results and the estimates are weighted by the average program size in the pre-treatment period. The standard errors follow from Porreca (2022). (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$)

Table A6: Main results, excluding international students

	New enrollments	Dropout after first year	Complete initial program	Years enrolled in initial program
ATT	-3.229 (3.449)	0.077*** (0.007)	-0.039*** (0.009)	-0.251*** (0.044)
N	7,525	7,525	7,525	7,525
	Obtain research university degree	Obtain higher education degree	Years enrolled in higher education	Annual earnings 12 years post enrollment
ATT	-0.015* (0.007)	-0.014* (0.006)	-0.061 (0.040)	317 (501)
N	7,525	7,525	7,525	6,482

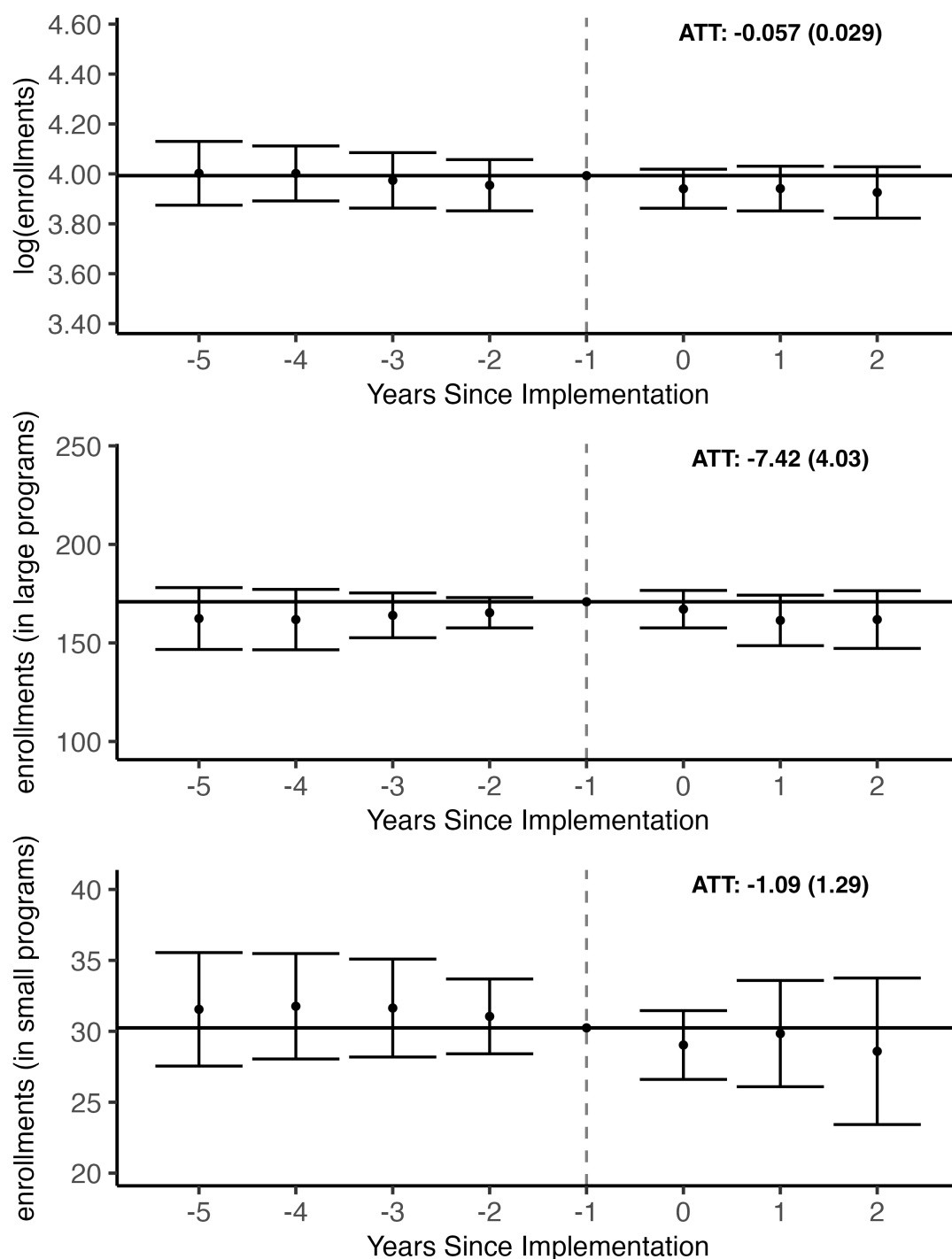
Notes: This table replicates the analysis from section 5. The only difference is that the sample here excludes all international students, who represent 11 percent of the core analysis sample. (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$)

Table A7: The effect of performance standards on antidepressant prescriptions

	Antidepressant prescription within two years of enrollment	Antidepressant prescription within five to eight years of enrollment
ATT	0.001 (0.002)	0.001 (0.003)
ATT ₋₁	0.001 (0.002)	0.003 (0.003)
ATT ₋₂	0.005 (0.003)	0.002 (0.003)
ATT ₋₃		0.000 (0.004)
ATT ₋₄		0.005 (0.004)
Mean	0.028	0.062
SD	(0.023)	(0.039)
Cohort	2006-2014	2002-2014
N	2,580	3,831

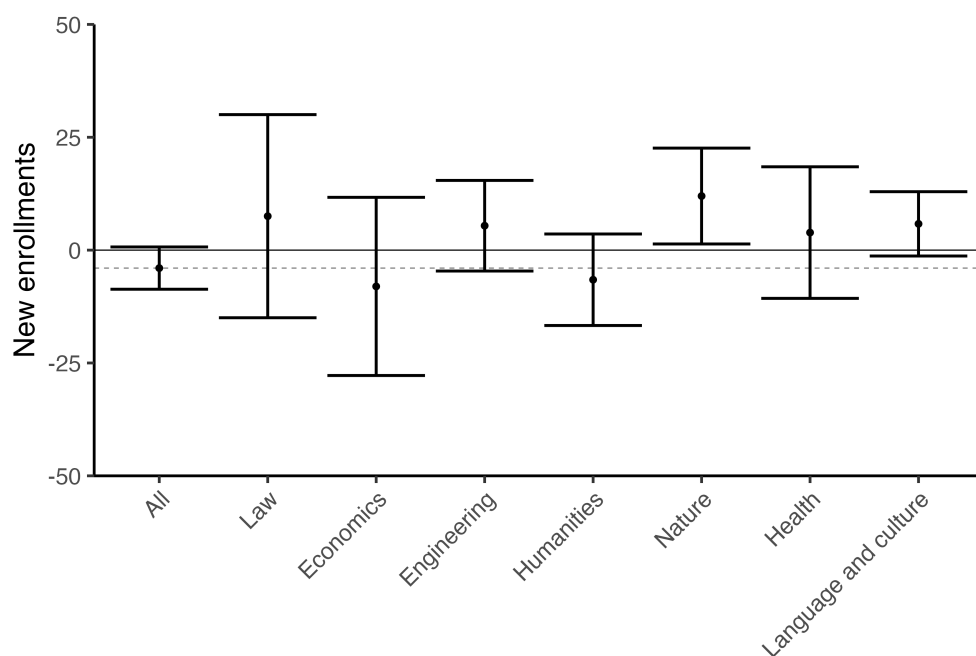
Notes: This table presents event-study estimates of the effect of implementing performance standards on an indicators for antidepressant within two years of the initial enrollment (column 1) and within five to eight years of the initial enrollment (column 2). Row 1 (ATT) presents the Average Treatment effect on the Treated, which is a weighted average of the ATTs for three post-treatment periods. The next four rows present placebo estimates based on pre-treatment period differences. The last two rows report the mean and standard deviation of each outcome in the last pre-treatment period. Standard errors are in parentheses. (***) $p < 0.001$, ** $p < 0.01$, * $p < 0.05$)

Figure A1: The effect of performance standards on new enrollments: in logs (all programs) and in levels (in large and small programs)



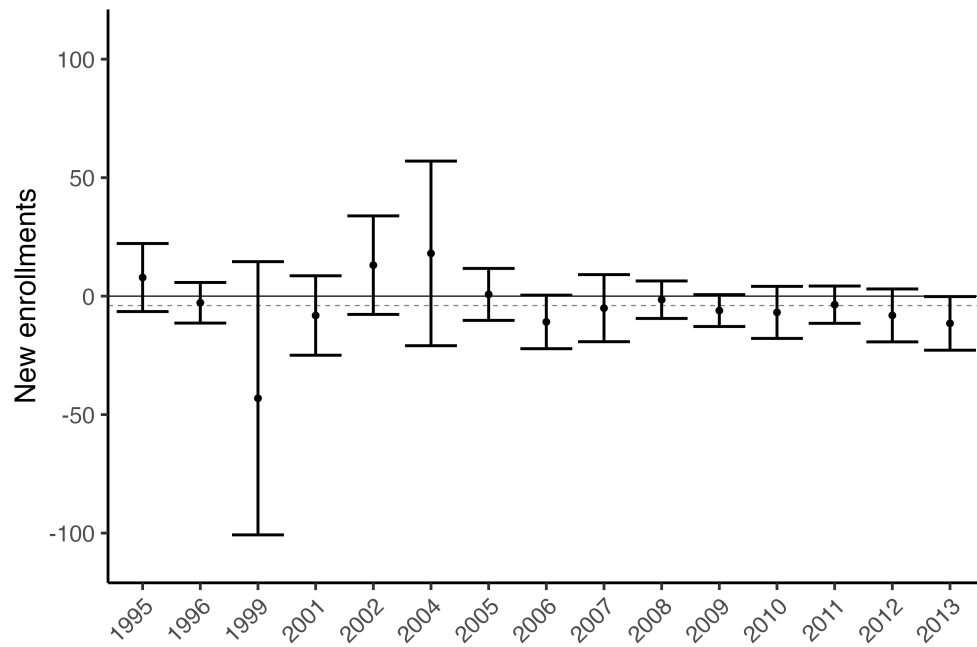
Note: Each panel presents event-study estimates of the effect of performance standards on new enrollments in logs (upper panel) or levels (middle and bottom panels). The upper panel uses all 7,555 observations (350 programs); the middle and bottom panels split the sample by above and below-median pre-treatment program size, respectively. Each graph is centered around the mean program size in the last pre-treatment period. Error bars are 95 percent confidence intervals.

Figure A2: Effects of implementing performance standards on program size across fields of study



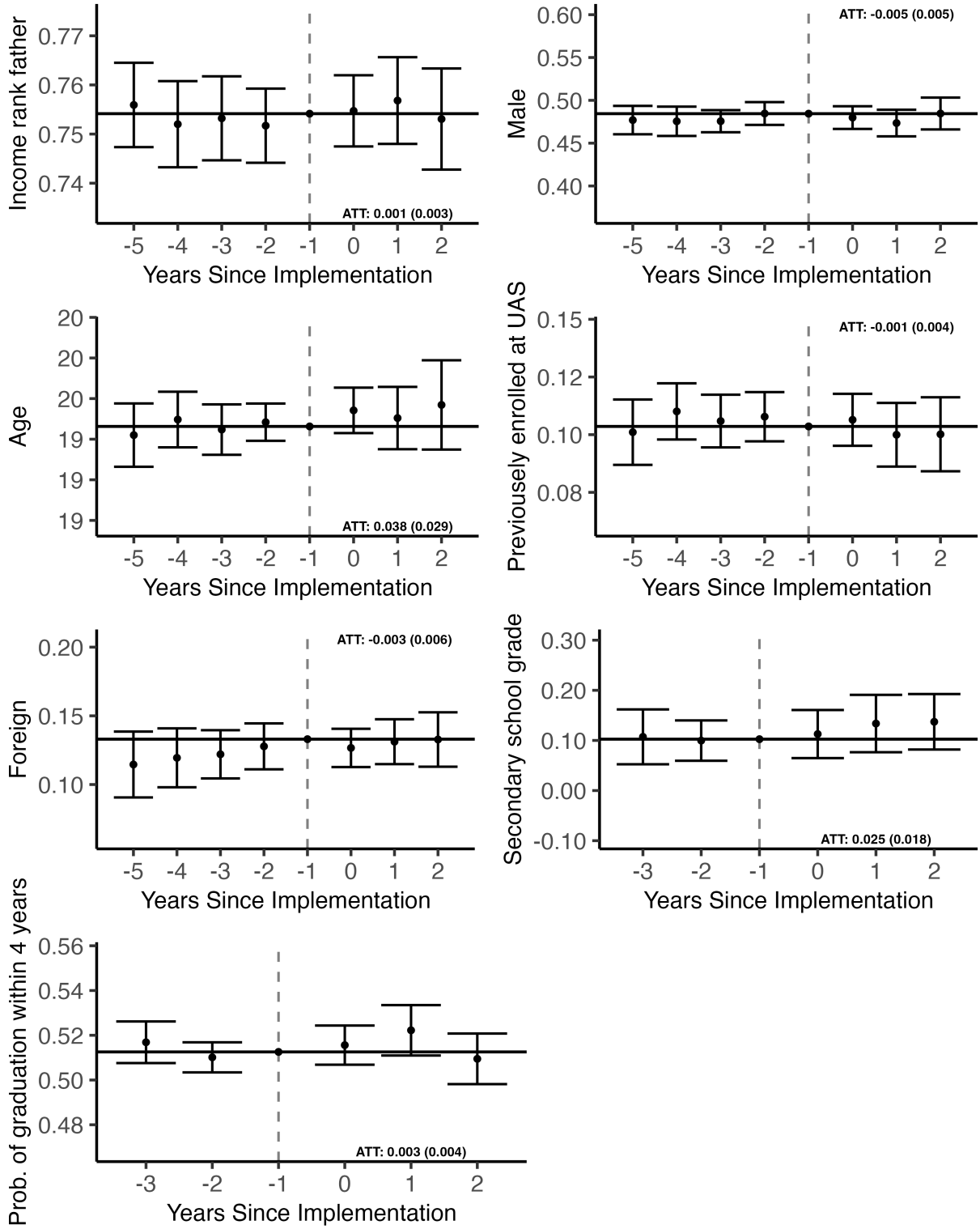
Note: This figure presents estimates of the effect of implementing performance standards on new enrollments across different fields of study. The estimates are obtained as follows: (i) select all treated and untreated observations for programs from a specific field and select all untreated observations from all other programs, (ii) apply difference-in-differences to this subsample to obtain the field-specific ATT, and (iii) repeat this process for all other fields. The dashed lines are centered around the ATT for the full sample. The error bars present 95% confidence intervals.

Figure A3: Effects of implementing performance standards on program size across implementation cohorts



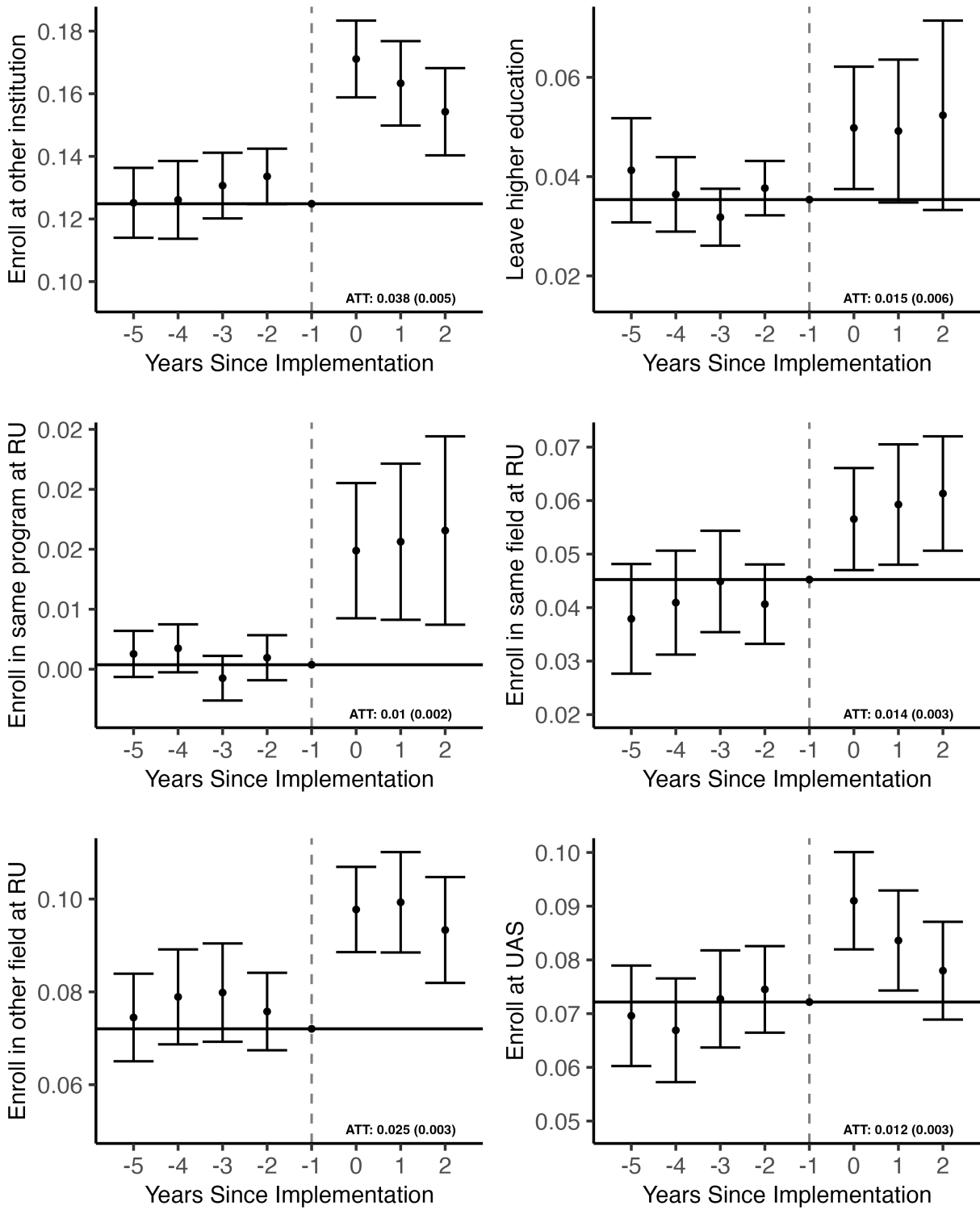
Note: This figure presents estimates of the effect of implementing performance standards on new enrollments across all adoption cohorts. The estimates are constructed using a group-level decomposition of the full sample ATT (Callaway and Sant'Anna (2021)). The dashed lines are centered around the ATT for the full sample. The error bars present 95% confidence intervals.

Figure A4: Event study estimates from Table 2



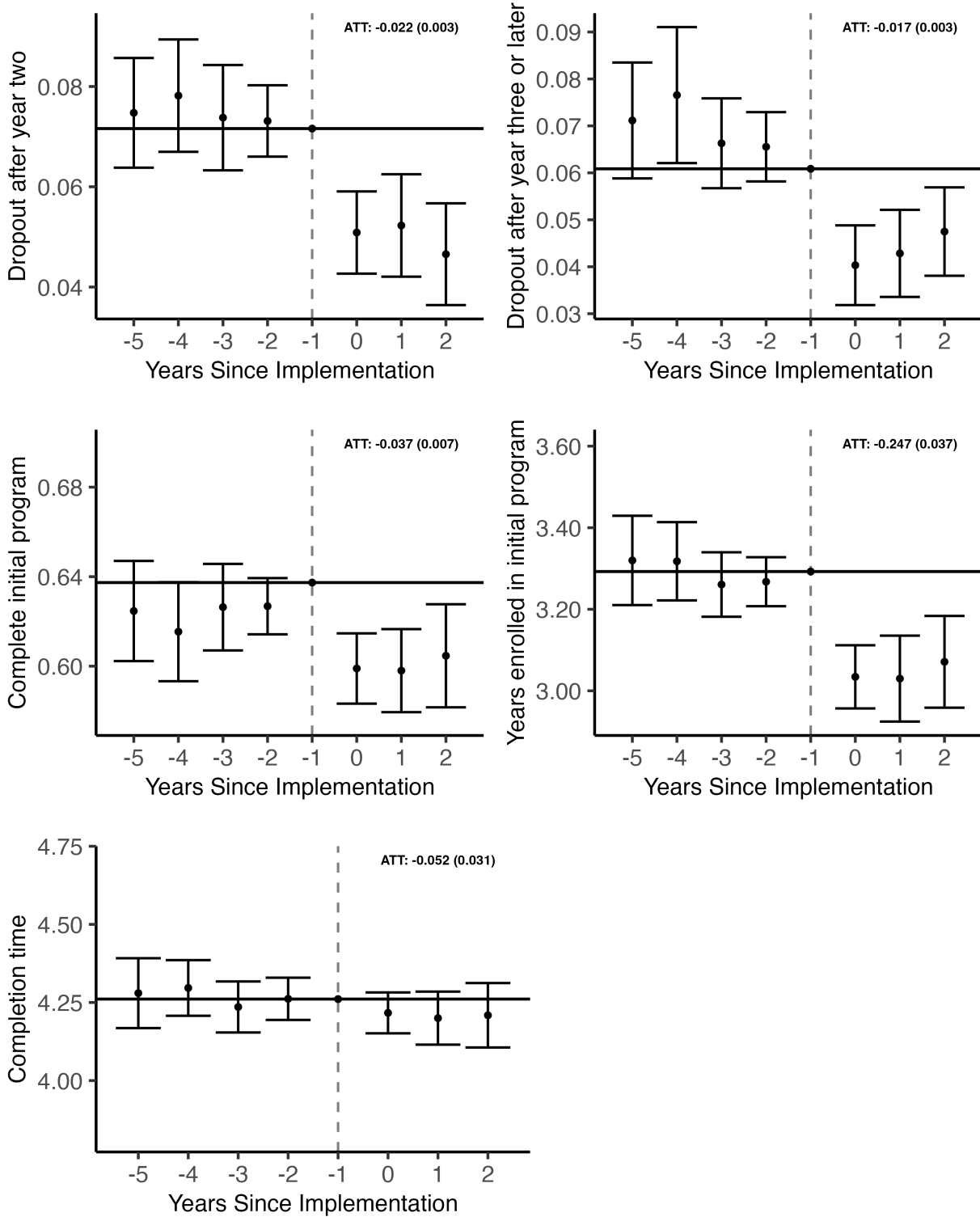
Note: This figure presents the event-study estimates corresponding to the results in Table 2. The graphs are centered around the average of the outcome in the last pre-treatment period. The error bars present 95% confidence intervals.

Figure A5: Event study estimates from Table 3



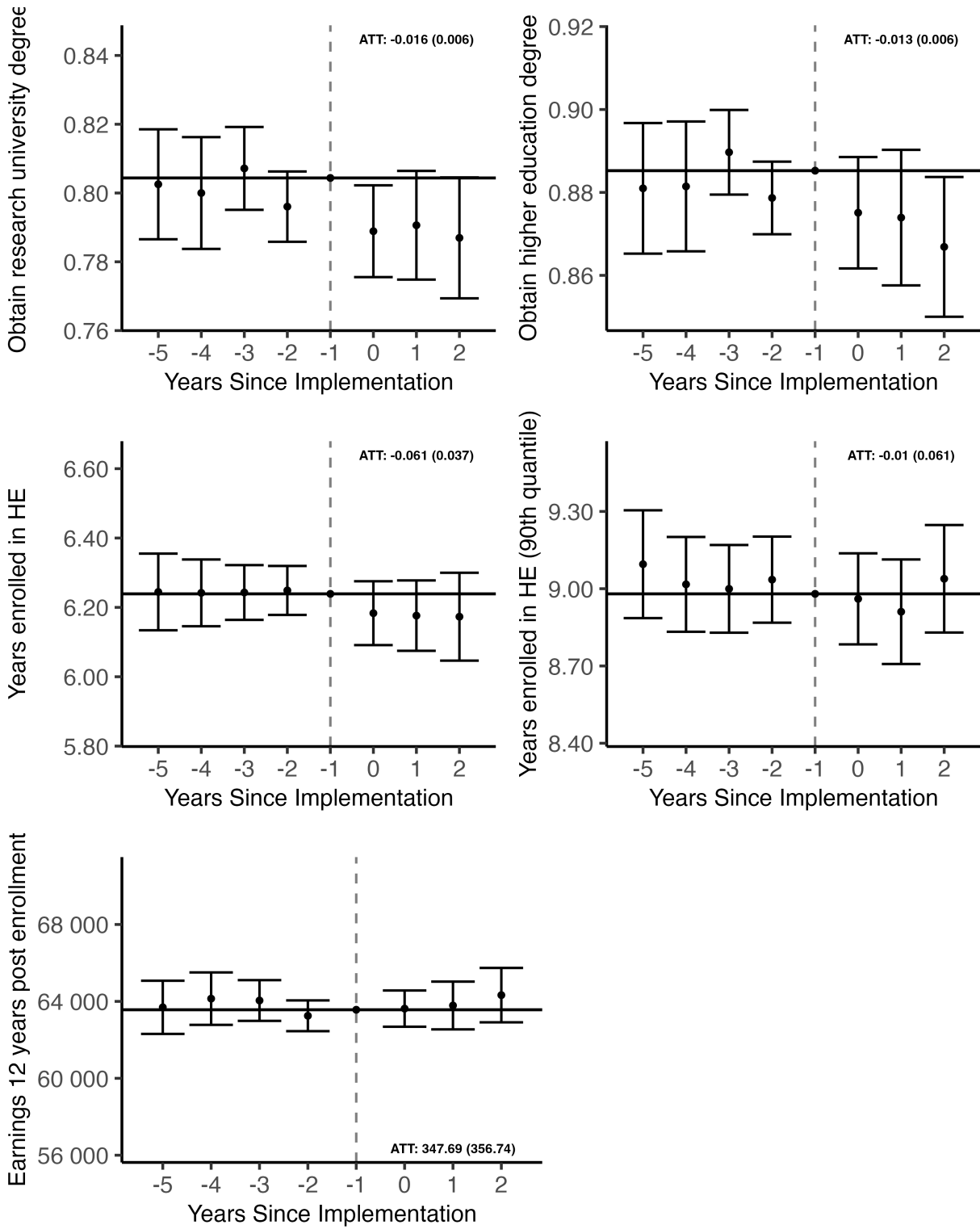
Note: This figure presents the event-study estimates corresponding to the results in Table 3. The graphs are centered around the average of the outcome in the last pre-treatment period. The error bars present 95% confidence intervals. RU stands for Research University. UAS stands for University of Applied Sciences.

Figure A6: Event study estimates from Table 4



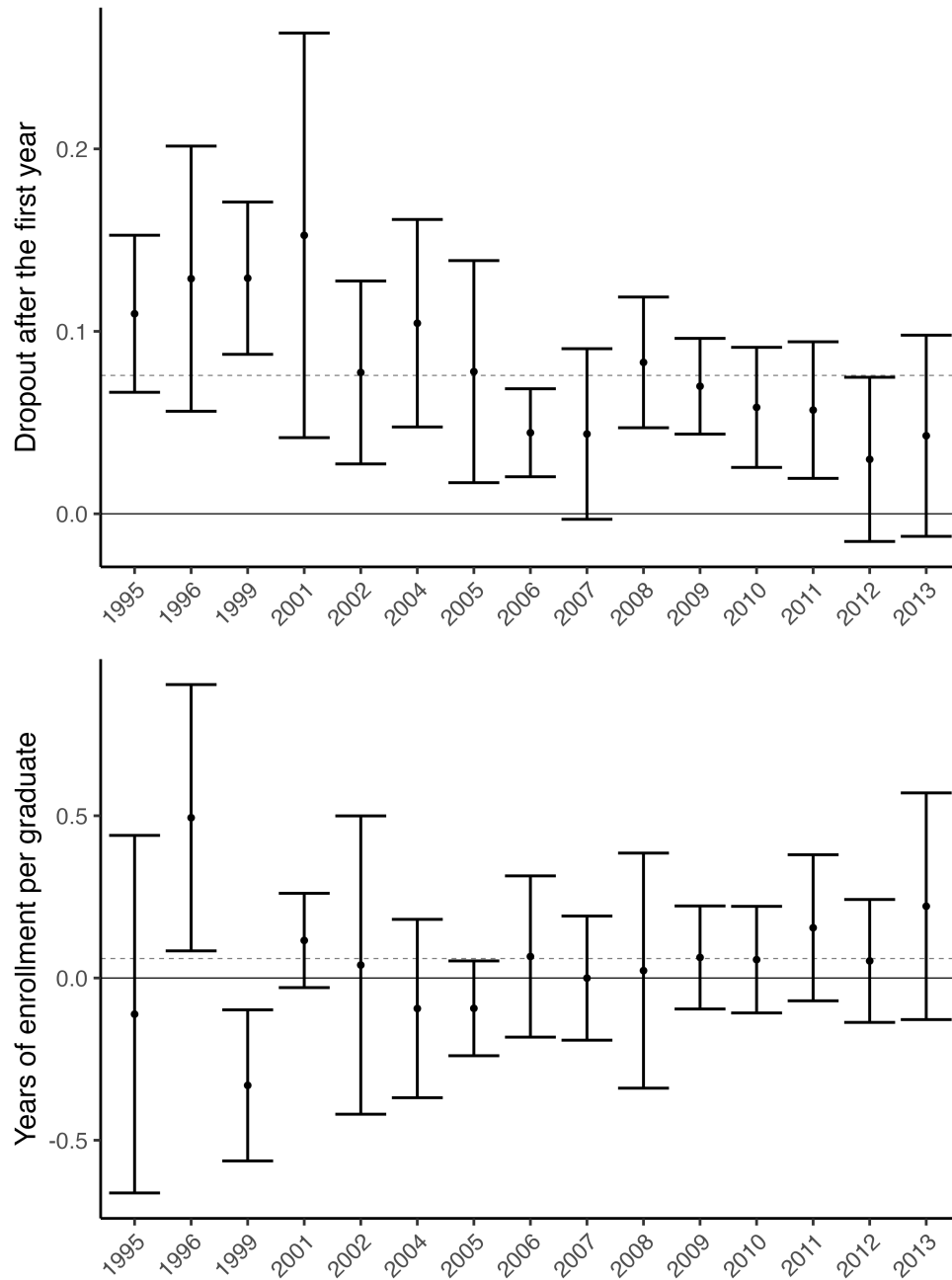
Note: This figure presents the event-study estimates corresponding to the results in Table 4. The graphs are centered around the average of the outcome in the last pre-treatment period. The error bars present 95% confidence intervals.

Figure A7: Event study estimates from Table 5



Note: This figure presents the event-study estimates corresponding to the results in Table 5. The graphs are centered around the average of the outcome in the last pre-treatment period. The error bars present 95% confidence intervals. HE stands for higher education.

Figure A8: Effects of implementing performance standards across adoption cohorts



Note: this figure presents estimates of the effect of implementing performance standards on first-year dropout (upper panel) and the average enrollment duration in higher education per graduate (lower panel) for all adoption cohorts. The estimates are constructed using a group-level decomposition of the full sample ATT. This methodology is explained in further detail in Callaway and Sant'Anna (2021). The dashed line is centered around the ATT for the full sample. The error bars present 95% confidence intervals.

Appendix B: Predicting Student Success

In Table 2 and Figure 4, I make use of students’ predicted field-specific on-time graduation chances. This section explains how these predictions are constructed.

I employ gradient-boosted decision trees to predict students’ chances of graduating within four years from their chosen program (Friedman (2001)). The explanatory variables include: gender, age, secondary school GPA, the highest income and education of the parents, migration generation, previously enrolled at a university of applied sciences (HBO), the level of high-school education, the field of study, and the subfield of study. The field of study and the subfield of study are based on classifications from the Ministry of Education. Missing values are replaced by indicators rather than excluded from the analysis. To ensure the independence of my predictions from the analysis sample, I train the models using only students who neither serve as the treatment nor control group. These consist of students in programs that implemented the BSA at least four years ago or will do so at least five years later. The sample only includes students enrolled after 2006 because grades are missing for earlier cohorts.

Table B1 reports the performance of the prediction exercise using students from the core analysis sample who are not used for training the prediction model. Students are divided into four equally sized groups, ranging from the lowest to the highest predicted probability of on-time graduation. Columns 2 and 3 show that the predictions are accurate on average. The remaining columns indicate that students with low predicted probabilities are more often male, relatively old, have less educated parents, and have a lower secondary school GPA.

Table B1: Prediction results in the unseen analysis sample

Prediction quantile	Prediction ($\hat{y}$)	Graduation within four years (y)	Male	Age	Income parents	Education parents	Migration Genera- tion	Previously enrolled in HBO	Secondary school GPA
0-25	0.30	0.30	0.76	20.21	0.73	15.40	0.13	0.13	-0.84
26-50	0.44	0.44	0.55	19.47	0.74	15.64	0.12	0.14	-0.31
51-75	0.55	0.53	0.42	19.04	0.75	15.82	0.15	0.10	0.23
76-100	0.69	0.67	0.25	18.78	0.76	15.97	0.12	0.04	1.13

Notes: This table reports results from the trained machine learning model on the unseen analysis sample. The machine learning model used is a gradient boosted decision tree (XGBoost). The model is trained to predict the likelihood that a student graduates within four years from the initial program. The training data includes all students from the core sample who were enrolled in programs at least five years before or four years after this program implemented a performance standard. The model is tuned using five-fold cross-validation. I sort the samples into predicted graduation quartiles (column 1) and calculate their predicted (column 2) and realized graduation rates (column 3). The remaining columns show averages across the different quartile.

Appendix C: Deterrence Effects

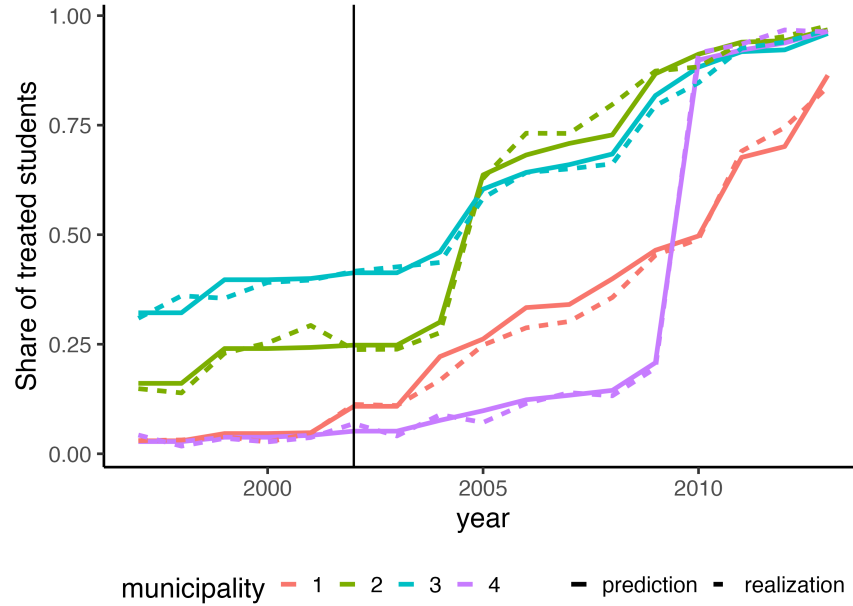
This appendix tests for deterrence effects using an alternative identification strategy. I show that prospective students who lived in the same municipality at age 16 choose similar programs over time, and that this pattern is unaffected by the introduction of performance standards. This suggests that performance standards neither deter university enrollment nor induce students to shift to other programs.

Specifically, I first select all students from the core sample for whom their municipality at age 16 is known (87.2 percent). For each municipality, I then calculate the share of students from this municipality that enrolled in each program between 1993 and 2002. For example, 1 percent of students from a given municipality could be enrolled in economics at the Erasmus University Rotterdam, 2 percent in mathematics at Leiden University, and so on. Based on these pre-2003 shares and the adoption rate of performance standards in these programs, I predict for each municipality how many students are expected to be in a program with a performance standard for each year after 2002. If students do not change their enrollment behavior, then these predicted shares should correspond to the observed shares.

To illustrate this approach, Figure C1 shows the predicted and realized share of students exposed to a performance standard for the four municipalities with the largest number of individuals who lived there at age 16 and later enrolled at a university. Consider municipality 2. Between 2002 and 2009, predicted exposure increases by about 20 percent, whereas in other municipalities, it rises more quickly. This is because only a few previously popular programs in municipality 2 implemented a performance standard during this period. In 2009, however, the predicted exposure jumps to almost 90 percent because many previously popular programs among students from this municipality implemented a performance standard that year. These programs are from a nearby university that adopted the performance standard in 2009. Crucially, the figure shows that the observed share of treated students closely mirrors the predicted share.

This also holds for the broader sample. To show this, I regress a student's observed treatment status after 2002 on their predicted treatment status based on the pre-2002 shares, including municipality and year fixed effects. These fixed effects ensure that variations in the predicted treatment are due to the staggered implementation of performance standards by previously popular programs. The coefficient equals 1.013 (Table C1), indicating a virtually perfect alignment between predicted and realized shares. This shows that whenever previously popular programs implemented a performance standard, the number of students exposed to these standards increased proportionally. This is inconsistent with deterrence effects.

Figure C1: Predicted and Realized Share of Treated Students in Four Municipalities



Note: the dashed lines show predicted shares of students enrolled in programs with a performance standard for the four municipalities with the largest student population. Predictions are constructed as follows: (i) select all first year students between 1993 and 2002 who lived in the municipality at age 16, (ii) calculate the share of these students in each program, and (iii) for a given year, sum the shares of students in programs that have a performance standard in that year. The solid lines show the observed shares. The vertical line at 2002 marks the last year used to construct predictions; all predicted shares after 2002 are therefore based solely on pre-2003 enrollments and evaluated against observations not used in generating the predictions.

Table C1: Testing for deterrence effects at the municipality level

	Observed treatment status
Predicted treatment status	1.013*** (0.014)
N	324,694
Municipality fixed effects	x
Year fixed effects	x

Notes: This table reports results from a regression of an indicator for enrollment in a program with a performance standard on the student's predicted likelihood of such enrollment. The specification includes fixed effects for municipality at age 16 and year of enrollment. Predicted treatment status is constructed as follows: (i) select all first year students between 1993 and 2002 who lived in the same municipality at age 16, (ii) calculate the share of these students in each program, and (iii) for a given year, sum the shares of students in programs that have a performance standard in that year. The estimation sample consists of all first year students in the core sample who enrolled after 2002 and for whom municipality at age 16 is observed. Restricting to post 2002 cohorts ensures that predictions are evaluated on observations not used to construct them. Standard errors are in parentheses. (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$

Appendix D: Survey evidence

This section describes the data collection, phrasing of questions, and the sample I use for the survey experiment.

The data come from a survey administered to first-year bachelor students from the public university Vrije Universiteit (VU) Amsterdam over a one-week period in November 2024. The students were recruited from a research participation course run by the behavioral lab of the School of Business and Economics. Students were informed that the study consisted of several questions related to their expectations and performance in the first year of the program.

The data were collected through an online survey (using Qualtrix). The survey took approximately 15 minutes to complete and consisted of several parts. Students received a small number of course credits upon the completion of the survey. I also included two attention checks and informed students that failing this results in withdrawal from the study. Of the 343 respondents, 10 failed the attention check, resulting in an overall sample of 333 students.

Subjective expectations. In addition to questions about demographics, personality traits, and secondary school performance, there are two main questions used for the analysis. The first question is aimed at eliciting students' expected number of course credits by the end of the first year. The specific question is:

At this moment, how many credits do you expect to have by the end of the first year? Divide a 100 points over the following options:

(30 or less, 31-36, 37-42, 43-48, 49-54, 55-60)

For instance, assigning 20 points to 31-36 credits implies a 20% chance of attaining between 31 and 36 credits by the end of the year. Assigning zero points signifies a belief that obtaining that specific number of credits is not possible.

The question has a built-in logical check, so that the percent chances of the number of course credits always sum to 100.

The performance threshold in these bachelor programs is 48 credits, and all courses yield 6 credits upon passing. I construct the subjective probability of dismissal by summing the probabilities of receiving 30 credits or less, between 31 and 36 credits, and between 37 and 42 credits.

Willingness to forgo gifts. The second set of questions present the hypothetical choice scenarios. In order to maximize respondents' reliability, I first present the following explainer:

We will present you some hypothetical situations where you have to choose between two alternatives. It is important that your choices here are realistic. Hence, in each of the following choice tasks, please consider carefully that you would actually prefer the alternative you choose.

After this, the main question reads:

Suppose you were given the choice between the following: receiving a payment today or an immediate removal of the Binding Study Advice. This would imply that you may continue with the program next year, regardless of how many course credits you have by the end of the year. We will now present to you five situations. The payment is different in every situation. For each of these situations we would like to know which you would choose.

Each of the following five questions is of the following form:

Would you rather receive a gift of X Euro today or an immediate removal of the Binding Study Advice?

The first question always corresponds to $X = 300$. Whenever a student chooses ‘Gift’, then the next question decreases the gift value, and if the student chooses for the removal of the BSA, then the next question increases the gift value. The values of X vary according to the scheme in Figure D1.

There are two special cases, indicated by $X = 0$ and $X = -30$ in Figure D1. If students choose a gift of 25 euros over the removal of the performance standard, then they get the following question (denoted by $X = 0$ in Figure D1): *What would you prefer: ‘**keeping** the Binding Study Advice (BSA) over its immediate removal?’* If a student chooses ‘Keeping the BSA’, then the next question is (denoted by $X = -30$ in Figure D1): *Would you prefer **paying** 30 euro today or an immediate removal of the Binding Study Advice?* If the student preferred the removal of the BSA over keeping the BSA, then the next question corresponds to $X = 10$.

Figure D1: Staircase method scheme

